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Channel Knowledge Map Prediction with Deep Learning for 6G UAV-Enabled Networks

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of the requirements for the
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Resum

Aquest TFG estudia la prediccio de mapes de coneixement de canal per a xarxes 6G amb UAVs. A partir de mapes urbans de 513 x 513 pixels, mascares LoS/NLoS i altura de l'antena, es prediuen perdua de trajecte, dispersio temporal i dispersio angular. Despres de provar U-Net, cGANs, experts i priors fisics, el treball convergeix en un model hibrid: priors interpretables calibrats per Try 78 i Try 79, mes un model residual conjunt a Try 80. Els priors finals assoleixen 1.92 dB en perdua de trajecte, 28.10 ns en delay spread i 13.74 graus en angular spread. Els resultats finals de Try 80 queden pendents de substituir quan finalitzi l'entrenament.

Resumen

Este TFG estudia la prediccion de mapas de conocimiento de canal para redes 6G con UAVs. A partir de mapas urbanos de 513 x 513 pixeles, mascaras LoS/NLoS y altura de antena, se predicen perdida de trayecto, dispersion temporal y dispersion angular. Tras probar U-Net, cGANs, expertos y priors fisicos, el trabajo converge en un modelo hibrido: priors interpretables calibrados en Try 78 y Try 79, mas un modelo residual conjunto en Try 80. Los priors finales alcanzan 1.92 dB en perdida de trayecto, 28.10 ns en delay spread y 13.74 grados en angular spread. Los resultados finales de Try 80 quedan pendientes de sustitucion cuando termine el entrenamiento.

Summary

This thesis studies channel knowledge map prediction for 6G UAV-enabled networks. From 513 x 513 urban maps, LoS/NLoS masks, and antenna height, the project predicts path loss, delay spread, and angular spread. After testing U-Nets, cGANs, expert models, and physical priors, the final direction becomes a hybrid one: calibrated interpretable priors from Try 78 and Try 79 plus a joint residual model in Try 80. The finished priors reach 1.92 dB path-loss RMSE, 28.10 ns delay-spread RMSE, and 13.74 degree angular-spread RMSE. Final Try 80 test results are intentionally left as placeholders until training and checkpoint selection are complete.

For all those who failed throughout their journey.

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Contents

Summary . . . 2
Acknowledgements . . . 4
Revision history and approval record . . . 5
Contents . . . 6
List of figures . . . 10
List of tables . . . 12
Abbreviations . . . 13

1 Introduction **15**
1.1 Work goals . . . 15
1.2 Requirements and specifications . . . 16
1.3 Methods and procedures . . . 16
1.4 Work plan . . . 17

2 State of the art **19**
2.1 Channel knowledge maps as the thesis context . . . 19
2.2 Classical propagation models and channel statistics . . . 20
2.2.1 Free-space and two-ray propagation . . . 20
2.2.2 Empirical and semi-empirical models . . . 20
2.2.3 UAV and air-to-ground channel models . . . 20
2.2.4 Delay spread and angular spread in standard channel models . . . 21
2.3 Deep learning for radio-map estimation . . . 21
2.3.1 Image-to-image translation baselines . . . 21
2.3.2 PMNet and the ICASSP outdoor challenge family . . . 22
2.3.3 Transformers, state-space models, and attention-based encoders . . . 23
2.3.4 Foundation models for radio maps . . . 23
2.4 Physics-informed and hybrid neural approaches . . . 23
2.4.1 Geometry-assisted and diffraction-aware deep learning . . . 23
2.4.2 Physics-informed neural networks . . . 24
2.4.3 Prior-residual decomposition . . . 24
2.5 Channel statistics prediction: delay spread and angular spread . . . 24
2.5.1 Empirical models for spread parameters . . . 24
2.5.2 Deep learning for wideband channel statistics . . . 25
2.5.3 Distributional mismatch and the spike-plus-exponential problem 25

2.6 Distribution-aware and multi-target modelling . . . 25
2.6.1 Heteroscedastic uncertainty . . . 25
2.6.2 Gaussian mixture model heads . . . 26
2.6.3 Stochastic diffusion models . . . 26
2.6.4 Multitask and joint prediction . . . 26
2.7 Training stabilisation and regularisation techniques . . . 27
2.7.1 Stochastic weight averaging . . . 27
2.7.2 Exponential moving average weights . . . 27
2.7.3 Feature-wise linear modulation . . . 27
2.8 Key benchmarks and fair-comparison protocol . . . 27
2.8.1 Why headline numbers cannot be compared directly . . . 27
2.8.2 Tarhouni et al. — measured suburban path-loss prediction . . . 28
2.8.3 AIRMap — adaptive digital-twin radio maps . . . 29
2.8.4 Gao et al. — multi-layer segmentation with corridor weighting . . . 29
2.8.5 PathFinder — multi-transmitter domain adaptation . . . 29
2.8.6 ICASSP 2025 indoor challenge . . . 30
2.9 Emerging techniques for closing the NLoS gap . . . 30
2.10 Position of Try 80 in the literature . . . 31
2.10.1 What is novel about Try 80 . . . 31
2.10.2 What Try 80 does not claim . . . 31
2.10.3 Summary of position . . . 32

3 Methodology / project development **33**
3.1 Dataset protocol . . . 33
3.2 From U-Net and cGAN to masked baselines . . . 33
3.3 Chronological attempt analysis . . . 34
3.4 Physical priors and PMHHNet . . . 34
3.4.1 Mathematical Evolution: From PMNet to PMHHNet . . . 35
3.4.2 Limitations and Legacy . . . 35
3.5 Distribution-first modelling . . . 36
3.6 Try 78 and Try 79 priors: overview and literature basis . . . 36
3.6.1 High-level architecture . . . 38
3.6.2 What was reused from the literature, what was changed, and what is new . . . 39
3.7 Try 78: Path Loss Prior . . . 41
3.7.1 LoS branch: coherent two-ray model . . . 41
3.7.2 Calibration of the LoS two-ray model . . . 45
3.7.3 NLoS branch . . . 48
3.7.4 Evaluation metric: pixel-weighted RMSE . . . 58
3.7.5 Observed Try 78 results . . . 58
3.8 Try 79: Delay and Angular Spread Priors . . . 61
3.8.1 Physical motivation for log-domain modeling . . . 61
3.8.2 Pixel-wise notation for the Try 79 fit . . . 61
3.8.3 Raw spread prior in log domain . . . 61

3.8.4 Calibrated spread prior . . . 62
3.8.5 Try 79 results . . . 67
3.9 Try 80: Joint Prior-Anchored Residual Model . . . 68
3.9.1 Experimental protocol and reporting status . . . 68
3.9.2 Prior-Anchored GMM Residual Model . . . 69
3.9.3 Architecture: Shared UNet with FiLM . . . 71
3.9.4 Composite Optimization Objective . . . 73

4 Results **78**
4.1 Experiments and Tests . . . 78
4.2 RMSE evolution and interpretation . . . 78
4.3 Data Visualization . . . 80
4.4 Limitations . . . 81

5 Sustainability Analysis and Ethical Implications **85**
5.1 Sustainability matrix . . . 85
5.1.1 Environmental impact . . . 86
5.1.2 Economic impact . . . 86
5.1.3 Social impact . . . 87
5.2 Ethical implications . . . 87
5.3 Relationship with the Sustainable Development Goals . . . 88

6 Conclusions and Future Work **89**
6.1 Conclusions . . . 89
6.2 Future Directions . . . 89

Bibliography **91**

A Reproducibility Appendix **95**
A.1 Detailed attempt log . . . 95
A.1.1 Early image-to-image phase . . . 95
A.1.2 Artifact removal and masked supervision . . . 95
A.1.3 Prior-guided residual phase . . . 96
A.1.4 Final distribution and prior phase . . . 96
A.2 Try 80 configuration summary . . . 97
A.3 Where the formulas are placed . . . 97
A.4 Final-result placeholder . . . 98

List of Figures

1.1 Project's Gantt diagram 18

3.1 Try 76 distribution-first path-loss system. Stage A learns global GMM parameters (π_k, μ_k, σ_k) from pooled features conditioned on UAV height; Stage B produces per-pixel soft assignments and within-component offsets anchored to those parameters. 37

3.3 Overall `Try 80` pipeline. The frozen priors are injected into the DNN, which learns a bounded correction δ around them. 39

3.4 Two-ray geometry. Direct path (blue) has length d_{LoS}; reflected path (green) has the same total length as the image-source path (gray dashed), which is d_{ref}. 42

3.5 Effective ground-reflection amplitude ρ^* as a function of UAV height, fitted independently per 5 m height bin. Values are stable in the range 0.47–0.56 for 10–130 m, then rise slightly at higher altitudes where training data is sparser. 48

3.6 Schematic comparison of FSPL and the coherent two-ray model for a 60 m UAV at 3.5 GHz ($h_{\mathrm{rx}} = 1.5$ m). The two-ray model produces a characteristic oscillatory pattern around the FSPL baseline, corresponding to alternating zones of constructive and destructive interference. 48

3.7 Try 78 pixel-weighted RMSE values. The coherent two-ray model reduces LoS RMSE from 3.95 dB (FSPL) to 1.75 dB, a gain of 2.2 dB. 59

3.8 Pipeline for the `Try 78` hybrid path loss prior. The LoS branch relies on an explicit two-ray interference model, while the NLoS branch calibrates a raw prior against map-derived morphology features using regime-wise OLS. The two branches are hard-switched based on the LoS mask to produce the final frozen prior. 60

3.9 Feature grouping used by the `Try 79` calibrated spread prior. The calibrated model is linear in $\mathbf{x}_{79}$, but the features include nonlinear transforms and interactions selected from propagation intuition. 64

3.10 Pipeline for the `Try 79` delay and angular spread priors. Both spread modalities share this identical linear architecture, predicting expected spread in the log domain via regime-wise ridge regression over 23 spatial and geometric features. 68

3.11 Schematic of the `Try 80` Shared UNet architecture with FiLM conditioning. 72

3.12 Detailed `Try 80` sinusoidal height-conditioning path. 72
3.13 Detailed prior-anchored GMM head used by `Try 80`. 73
3.14 Composite `Try 80` loss. The probabilistic terms train the mixture density, while the anchor and guard terms enforce the prior-residual interpretation. 75
3.2 Try 77 spread-prediction system. The Stage A mixture head separately models the near-zero spike (LoS direct path) and the heavy-tailed components (NLoS scattering). Stage B decodes these into spatially varying per-pixel predictions. 77

4.1 Current Try 80 validation snapshot against its frozen priors. This figure is a placeholder until final test evaluation is complete. 80
4.2 Example Try 79 inspection panel for a Vancouver sample. 80
4.3 Try 80 validation history against frozen priors. Placeholder diagnostic, not a final test result. 81

List of Tables

1.1 Main CKM dataset fields used in the thesis. 16

2.1 Corrected physical RMSE for ICASSP 2023 outdoor challenge methods. The normalised scores are taken from the challenge family and converted to dB using the 36 dB RadioMap3DSeer dynamic range described in the dataset paper [39, 38]. 22
2.2 Evaluation condition comparison across representative methods. 28
2.3 Emerging techniques ranked by expected impact on CKM NLoS RMSE. . 31
2.4 Positioning of Try 80 against the most relevant published methods. . . . 32

3.2 One-page summary of what is inherited from the literature, what is adapted to CKM, and what is reused or newly assembled in `Try 80`. . . . 38
3.3 Representative fitted LoS two-ray parameters by height bin. 47
3.4 Term-by-term inventory of the `Try 78` calibration vector. 52
3.5 Try 78 pixel-weighted RMSE (dB) on the evaluation split. The "Role" column indicates whether the model is used as a frozen prior in `Try 80`, serves only as an ablation baseline, or is an intermediate construction step. 58
3.6 Term-by-term inventory of the direct terms in the `Try 79` spread calibration vector. 63
3.7 Interaction terms in the `Try 79` spread calibration vector. 64
3.8 Delay spread: calibrated test RMSE (ns) by topology class. 67
3.9 Angular spread: calibrated test RMSE (°) by topology class. 67
3.1 Evolution of modelling techniques and lessons across the attempt history. 76

4.1 Representative experimental milestones. 79
4.2 Representative RMSE evolution across the project. Values are reported under their own experiment contract and are used to show trend and diagnosis, not to claim every row is directly comparable. 83
4.3 Try 78 final hybrid path-loss prior, pixel-weighted RMSE. 84
4.4 Try 79 calibrated test results, pixel-weighted RMSE. 84
4.5 Try 80 current validation snapshot. These are placeholders, not final test results. 84

4.6 Approximate quantisation RMSE implied by the stored target precision. The float16 values use the unit in the upper part of the observed range (about 180 for path loss/angular spread and 678 for delay spread); float32 is effectively negligible at this scale. 84

5.1 Sustainability matrix summary for the thesis. 85
5.2 Indicative economic cost of the BT development. 87

A.1 Main Try 80 configuration values. 97
A.2 Location of the main mathematical definitions. 98

Abbreviations

6G sixth-generation mobile network

A2G air-to-ground

ABS aerial base station

AS angular spread

BT bachelor thesis

cGAN conditional generative adversarial network

CGM channel gain map

CKM channel knowledge map

CNN convolutional neural network

CSI channel state information

DL deep learning

DS delay spread

EMA exponential moving average

ETSETB Escola Tecnica Superior d'Enginyeria de Telecomunicacio de Barcelona

EU European Union

FiLM feature-wise linear modulation

FSPL free-space path loss

GAN generative adversarial network

GEF Grau en Enginyeria Fisica

GMM Gaussian mixture model

GPU graphics processing unit

GREELEC Grau en Enginyeria Electronica de Telecomunicacio

GRETST Grau en Enginyeria de Tecnologies i Serveis de Telecomunicacio

HDF5 Hierarchical Data Format version 5

HPC high-performance computing

KL Kullback-Leibler divergence

LoS line of sight

LSF large-scale fading

MAE mean absolute error

ML machine learning

mmWave millimetre wave

NLL negative log-likelihood

NLoS non-line of sight

OLS ordinary least squares

PLE path-loss exponent

PMHHNet path-loss map network with high-frequency and height awareness

PMNet path-loss map prediction network

RMSE root mean squared error

Rx receiver

SDG Sustainable Development Goal

SLURM Simple Linux Utility for Resource Management

SOA state of the art

SWA stochastic weight averaging

TTA test-time augmentation

Tx transmitter

U-Net U-shaped convolutional encoder-decoder network

UAV unmanned aerial vehicle

CHAPTER 1

Introduction

Reliable radio coverage prediction is a central requirement for future 6G systems in which UAVs may act as aerial base stations, relays, or temporary coverage nodes. A classical ray-tracing simulator can produce detailed channel knowledge maps, but this is too expensive to repeat every time the transmitter height, the city morphology, or the deployment assumptions change. In the 6G literature, a channel knowledge map is a site-specific, location-tagged repository of channel information that can support environment-aware planning and reduce online channel-acquisition overhead [40, 41]. This thesis studies whether a learned surrogate can predict dense channel maps directly from urban geometry while preserving enough physical structure to generalize to cities not seen during training.

The project is formulated as dense image-to-image regression on 513×513 pixel city maps. The input is primarily a building-height topology map, a LoS/NLoS mask, and the UAV antenna height. The outputs studied through the project are path loss, delay spread, and angular spread. The final line of work combines interpretable non-deep-learning priors with deep residual models, so the neural network learns corrections around a strong physical anchor instead of rediscovering the whole propagation law from scratch.

1.1 Work goals

The initial technical targets were defined in physical units and were kept as the main success criteria throughout the project:

- path-loss RMSE below 5 dB;
- delay-spread RMSE below 50 ns;
- angular-spread RMSE below 20°;
- separate reporting for global, LoS, and NLoS regions.

The most important methodological objective was not only to obtain a low average RMSE, but to do so under a strict city-holdout protocol. In this setting, complete cities

are reserved for validation and testing. This is a harder and more deployment-relevant condition than random pixel or random tile splits, because the model must extrapolate to urban layouts it has never seen.

1.2 Requirements and specifications

The project uses the CKM HDF5 dataset `CKM_Dataset_270326.h5`. The canonical dataset contains 66 real-city scenarios and 16180 samples. Each sample is a 513×513 pixel raster with the following fields:

Table 1.1: *Main CKM dataset fields used in the thesis.*

Field	Shape	Type	Role
`topology_map`	513^2	float32	Building height in metres. Pixels with buildings are not valid receiver locations.
`los_mask`	513^2	uint8	Binary LoS/NLoS visibility mask.
`path_loss`	513^2	uint8	Ray-traced path loss in dB, quantized at approximately 1 dB.
`delay_spread`	513^2	uint16	Delay-spread target, stored as integers and evaluated in ns after decoding.
`angular_spread`	513^2	uint8	Angular-spread target, stored as integers and evaluated in degrees after decoding.
`uav_height`	scalar	float32	UAV transmitter height, used for conditioning and physical priors.

All losses and metrics are computed only on ground pixels. Building pixels are kept in the input topology because they explain propagation, but they are excluded from error accumulation because the receiver is assumed to be on the ground. The fixed receiver height used in the analytic priors is 1.5 m. The transmitter is the UAV antenna, whose height varies continuously across the dataset.

1.3 Methods and procedures

The work started from a U-Net and pix2pix-style cGAN baseline [12]. This was a natural first approach because the task is dense map prediction, but the adversarial component did not solve the physical errors that mattered most. Later attempts changed decoder upsampling, added multi-scale supervision, separated metrics and regimes, introduced physical prior channels, and finally moved to topology-partitioned experts.

The largest methodological change came after the models repeatedly failed on NLoS tails. Histogram analysis showed that the path-loss distribution in NLoS regions is not well

described by a single Gaussian-like target. This motivated Try 76: a distribution-first architecture with a sample-level GMM head followed by a map head that assigns pixels to distribution components. Try 77 applied the same idea to delay and angular spreads, where spike-plus-tail behaviour is even more visible.

The final stage changed the role of deep learning again. Try 78 built a strong interpretable path-loss prior from FSPL, coherent two-ray LoS propagation, and calibrated NLoS corrections. Try 79 built log-domain ridge-calibrated priors for delay and angular spread. Try 80 then combined these frozen priors with a shared multi-task residual network.

1.4 Work plan

The original work plan assumed a mostly linear path: literature review, dataset preparation, baseline U-Net/cGAN training, model improvement, evaluation, and writing. The actual project became more experimental because each stage exposed a different limitation. Figure 1.1 therefore shows the plan as the major technical phases rather than as all 80 individual attempts.

The main deviation from the initial plan was the discovery that a standard image-to-image CNN could look plausible in LoS-dominated regions while still failing badly in NLoS regions. This forced several changes: stricter ground-only masking, regime-level diagnostics, topology experts, physical priors, and finally the distribution-first and prior-anchored designs. The result is a project whose final method is more hybrid and more interpretable than the initial proposal anticipated.

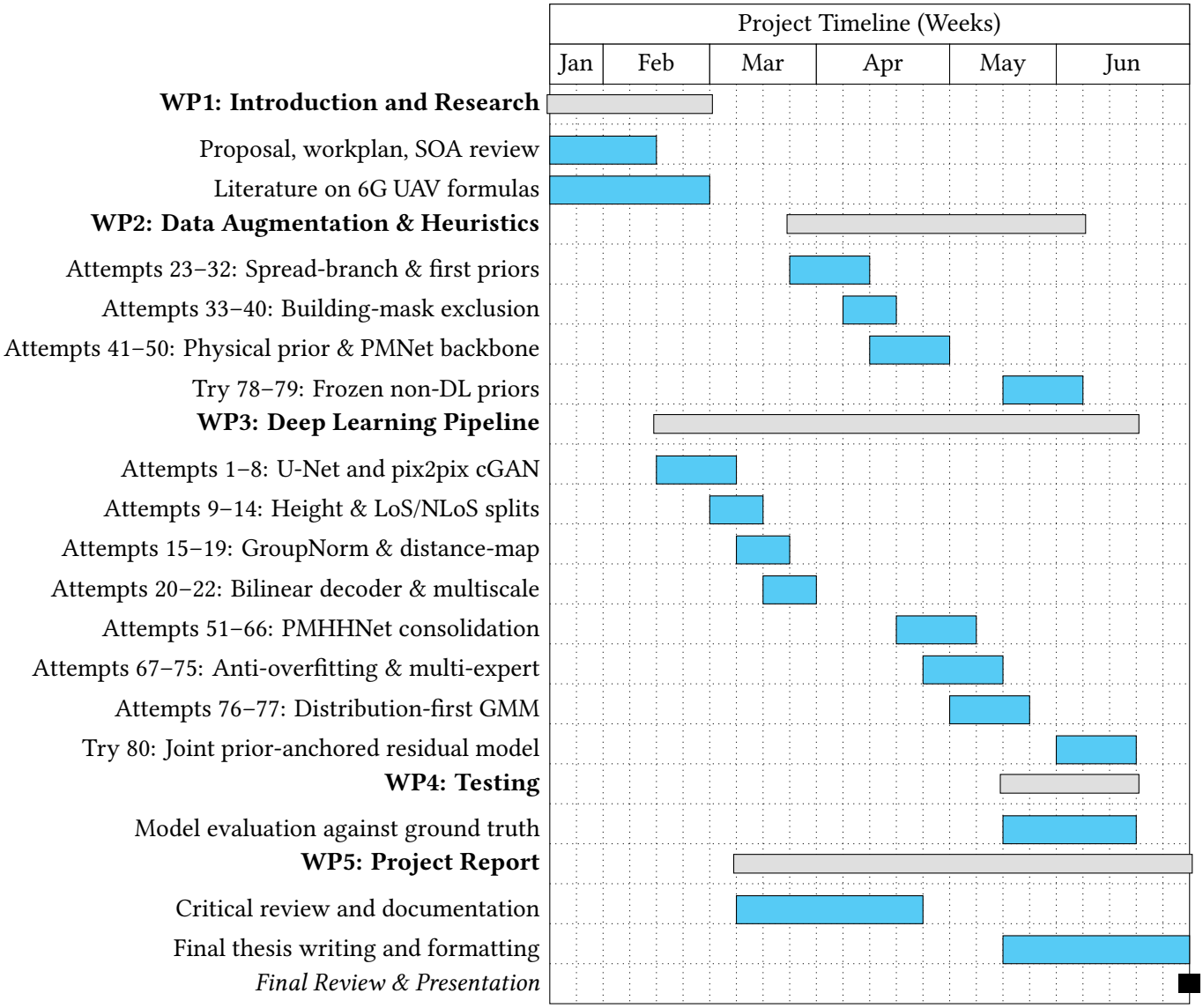

Figure 1.1: *Gantt diagram of the project, grouped by the main architectural phases rather than by individual attempts.*

CHAPTER 2

State of the art of the technology used or applied in this thesis

Radio-map prediction sits at the intersection of classical wireless propagation modelling, statistical channel analysis, and dense computer-vision regression. This chapter surveys the relevant state of the art across four axes that jointly motivate the design choices in Try 80: (i) classical and empirical propagation models, (ii) deep learning for radio-map estimation, (iii) physics-informed and hybrid neural approaches, and (iv) distributional and multi-target channel modelling. A final section frames fair-comparison criteria and positions Try 80 within the literature.

2.1 Channel knowledge maps as the thesis context

The term channel knowledge map (CKM) refers to a site-specific database of location-tagged channel knowledge, such as path loss, channel gain, angles, or delay information, used to make future radio systems environment-aware [40, 41]. This framing is more precise than a generic radio map: the map is not only an image-like coverage estimate, but a queryable prior that can reduce channel-state acquisition overhead, guide beamforming, support planning, and provide initial channel knowledge before online measurements are available.

Recent CKM literature also clarifies why the thesis uses a large offline ray-traced dataset rather than sparse field samples. Xu and Zeng [37] analyse how CKM construction error depends on spatial sample density and on the number of neighbouring samples used for online prediction, showing the explicit cost–accuracy trade-off of measurement-driven CKM construction. For a thesis whose goal is to learn dense 513×513 targets under city holdout, this motivates the simulation-first route: it provides full supervision everywhere, while the final model is still evaluated as a surrogate for a ray-traced CKM rather than as a replacement for field calibration.

The data ecosystem has also expanded since the first path-loss challenge. RadioMap3DSeer [38] and the ICASSP 2023 challenge [39] provide the closest open benchmark family for

path-loss image prediction, while the directive-antenna dataset of Jaensch et al. [15] and CKMImageNet [36] show the broader move toward open, location-specific datasets with richer channel and environment representations. The CKM dataset used in this thesis sits in the same data-driven tradition, but differs in its continuous UAV height conditioning, real-city holdout split, and simultaneous path-loss, delay-spread, and angular-spread targets.

2.2 Classical propagation models and channel statistics

2.2.1 Free-space and two-ray propagation

The most fundamental path-loss term is the *free-space path loss* (FSPL), which depends only on carrier frequency f and Tx–Rx distance d:

$$\mathrm{PL}_{\mathrm{FSPL}} = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right). \tag{2.1}$$

FSPL is the irreducible floor in unobstructed conditions. In aerial scenarios a specular ground reflection produces constructive/destructive interference periodically with distance. The *coherent two-ray* model captures this by adding the direct and reflected fields before converting to dB [27, 32]:

$$E_{\mathrm{total}} = E_{\mathrm{direct}} + \Gamma \cdot E_{\mathrm{reflected}} \cdot e^{j\Delta\phi}, \tag{2.2}$$

where Γ is the reflection coefficient and $\Delta\phi$ is the phase difference determined by the path-length difference. A calibrated two-ray model fitted to the CKM dataset (Try 78 of this thesis) achieves approximately 1.75 dB RMSE on line-of-sight (LoS) pixels — the deterministic radial structure explains most LoS variance without any neural network.

2.2.2 Empirical and semi-empirical models

Empirical models such as the Okumura–Hata and COST-231/Hata formulas extend FSPL by incorporating frequency, antenna heights, and city-type correction terms [27]. While computationally cheap, they are calibrated on specific frequency bands and morphology types, limiting accuracy when the geometry departs from the fitting dataset.

The 3GPP Technical Report 38.901 [1] formalises stochastic channel models for frequencies from 500 MHz to 100 GHz for urban macro (UMa), urban micro (UMi), rural macro (RMa), and indoor scenarios. It provides separate path-loss polynomials for LoS and non-LoS (NLoS) conditions, breakpoint distances, and shadow-fading standard deviations, which serve as reference baselines throughout this thesis.

2.2.3 UAV and air-to-ground channel models

Classical terrestrial channel models assume the transmitter is at rooftop height or below. Unmanned aerial vehicle (UAV) base-station scenarios introduce a third dimension:

the UAV can fly from near-ground level to several hundred metres. Al-Hourani et al. [10] showed that the LoS probability increases monotonically with elevation angle and depends on city morphology type (suburban, urban, dense urban, high-rise). The ITU and 3GPP A2G channel model family [1, 17] parameterises this dependence through logistic functions of the elevation angle $\theta = \arctan\big((h_{\mathrm{tx}} - h_{\mathrm{rx}})/d_{2D}\big)$, where h_{tx} and h_{rx} are transmit and receive heights respectively and d_{2D} is the horizontal distance.

Recent work by Saboor and Vinogradov [30] makes this height dependence especially relevant for the present thesis: in simulated 26 GHz urban A2G links, LoS path loss remains close to a free-space exponent, whereas NLoS path-loss exponents and shadow fading evolve with ABS height and with urban layout. Vinogradov et al. [34] also propose 3D shadow projections as a fast way to generate spatially consistent A2G LoS maps without full point-by-point ray tracing. These results support two design choices used throughout this work: treating the LoS/NLoS mask as a physical input rather than as a cosmetic label, and conditioning every dense prediction on UAV height.

A key finding from the CKM dataset is that LoS path-loss residuals after two-ray subtraction form almost perfect radial rings around the transmitter. This confirms that buildings act primarily as a *visibility mask* in LoS conditions rather than as the main source of residual path-loss variance, validating the prior-subtraction philosophy of Try 78.

2.2.4 Delay spread and angular spread in standard channel models

Beyond path loss, full channel characterisation requires second-order statistics. The *root-mean-square delay spread* τ_{rms} measures the time-domain dispersion of multipath arrivals and drives inter-symbol interference in wideband systems. The *azimuth angular spread* σ_{AS} at the receiver quantifies the spatial diversity of arriving paths.

3GPP TR 38.901 [1] models both quantities as log-normal random variables conditioned on the LoS/NLoS state and distance. The WINNER II and COST 2100 models adopt the same convention. In practice, the distributions of delay spread and angular spread are right-skewed with a heavy exponential tail and often include a spike near zero representing the dominant LoS path. Try 79 of this thesis empirically confirms this *spike-plus-exponential* shape for all topology classes in the CKM dataset, with skewness exceeding 5 and kurtosis exceeding 40 in several morphology classes. A Gaussian model cannot represent this shape faithfully, motivating the mixture and distributional approaches of Tries 77 and 80.

2.3 Deep learning for radio-map estimation

2.3.1 Image-to-image translation baselines

The radio-map prediction problem can be cast as an image-to-image regression: given an input topology map (and optionally a transmitter location channel), the model predicts a path-loss map of the same spatial extent. The pix2pix framework [12] established a general recipe for such tasks using a conditional GAN with a U-Net generator and

a PatchGAN discriminator. Early iterations of this thesis used this formulation. In practice the adversarial objective sharpens spatial texture but does not address the height-dependent radial pattern or the heavy NLoS tails that dominate the error budget in the CKM dataset.

2.3.2 PMNet and the ICASSP outdoor challenge family

PMNet [19] was the winner of the first ICASSP 2023 Outdoor Pathloss Radio Map Prediction Challenge [39]. It uses a deep encoder-decoder with skip connections, trained and evaluated on the RadioMap3DSeer dataset [38]. RadioMap3DSeer consists of procedurally generated city maps with transmitters fixed at rooftop height (6–20 m) and building interior pixels masked to zero error.

The challenge metric is RMSE in normalised image units. The RadioMap3DSeer image mapping clips path gain to the range $[\mathrm{PL}_{\mathrm{thr}}, \mathrm{PL}_{\mathrm{max}}] = [-111, -75]$ dB, yielding a 36 dB dynamic window. Converting to physical dB requires multiplying normalised RMSE by 36. After this correction, the true physical errors are:

Table 2.1: *Corrected physical RMSE for ICASSP 2023 outdoor challenge methods. The normalised scores are taken from the challenge family and converted to dB using the* 36 dB *RadioMap3DSeer dynamic range described in the dataset paper [39, 38].*

Method	RMSE (norm.)	True RMSE (dB)
RMTransformer [21]	0.0270	0.97
PMNet [19]	0.0383	1.38
PPNet baseline [39]	0.0507	1.83

These numbers are not directly comparable to the CKM evaluation for three structural reasons: (i) fixed Tx height eliminates the continuous UAV height-conditioning challenge; (ii) same-distribution train/test scenes allow implicit generator-pattern memorisation; (iii) the 36 dB window is much narrower than CKM's effective 65–130 dB range, intrinsically compressing RMSE values. The table is therefore useful as a scale reference for modern image-to-image radio-map predictors, not as evidence that the Try 80 task is solved by those architectures. In particular, Try 80 must predict over unseen real-city geometries, at native 513×513 resolution, with a continuously varying UAV altitude and with metrics computed only over valid ground receiver pixels.

A separate transfer test performed in this thesis (the "weird tries" baseline) placed official PMNet checkpoints on CKM samples. The best zero-shot result was approximately 84 dB RMSE, with most models collapsing to a nearly constant prediction. After oracle linear calibration from test data the minimum achievable RMSE was still approximately 42 dB. This confirms that PMNet's strong challenge score does not transfer to the structurally harder CKM setting.

2.3.3 Transformers, state-space models, and attention-based encoders

The RMTransformer [21] replaced PMNet's convolutional encoder with a multi-scale attention mechanism, improving the normalised RMSE on RadioMap3DSeer to 0.0270 (true physical: 0.97 dB). The key advantage is a larger receptive field for capturing long-range shadowing correlations across wide urban streets.

RadioMamba and similar state-space sequence models applied to radio maps [35, 29] offer linear complexity attention. These architectures are effective when global context is necessary, such as tracking a shadowing corridor across an entire 513×513 map, but they require substantially more data and compute than convolutional methods to reach the same local detail quality.

2.3.4 Foundation models for radio maps

The most recent shift in the literature is towards *wireless channel foundation models* pre-trained on large heterogeneous datasets before task-specific fine-tuning. WiCo-PG [35] uses dual VQGANs and a transformer with frequency-guided mixture-of-experts. It exploits RGB drone camera images as an auxiliary modality and achieves an NMSE of 0.012 on its proprietary evaluation benchmark. RadioLAM [25] targets fine-grained 3D radio map estimation with a large-model architecture at millisecond inference latency. FM-RME [29] frames radio map estimation as a foundation modelling problem with self-supervised pre-training on simulated propagation data.

These foundation approaches represent the long-term direction of the field but require massive training corpora and compute budgets well beyond the scope of a single CKM dataset. They are cited here as the horizon against which this thesis positions its own contributions.

2.4 Physics-informed and hybrid neural approaches

2.4.1 Geometry-assisted and diffraction-aware deep learning

Purely data-driven models must implicitly learn diffraction geometry from the topology map. An alternative is to pre-compute a physics-based channel prior and provide it as an additional input channel, reducing the learning problem to residual estimation.

Shadow-projection methods [34] occupy an intermediate point between stochastic LoS-probability models and full ray tracing. They use 3D building geometry to project blocked ground regions from the aerial transmitter, producing a deterministic LoS map that is spatially consistent along user trajectories. The CKM dataset already provides a ray-traced LoS mask, so this thesis does not need to generate the mask itself; nevertheless, the method is directly relevant because it explains why a binary visibility channel can carry so much of the structure that the neural network otherwise has to infer from the height map alone.

Knife-edge diffraction [7] is a deterministic approximation to single-edge scattering based on ITU-R P.526 geometry. For each pixel, the first Fresnel zone obstruction height is computed from the 2D building silhouette and converted to a diffraction gain. Providing this as an extra input channel was shown to yield up to 18 % accuracy improvement in geometry-assisted deep learning experiments. Try 67 of this thesis implements this channel through the `knife_edge.py` module, feeding it into the PMHHNet encoder alongside the topology map.

2.4.2 Physics-informed neural networks

The ReVeal architecture [28] operates as a physics-informed neural network (PINN) by incorporating the Helmholtz residual as an auxiliary loss term. This penalises predictions that violate the wave equation, constraining the network to produce physically smooth field maps and reducing corner-transition artefacts. ReVeal achieves 1.95 dB RMSE in rural scenarios using only 30 training points by leveraging the physical constraint as a strong regulariser. Tries 67–69 of this thesis implement a lightweight analogue: a masked Laplacian penalty on the residual prediction, which penalises locally non-smooth outputs without solving the full Helmholtz operator.

2.4.3 Prior-residual decomposition

A principled hybrid approach is to *freeze* a calibrated analytic prior and train the neural component to predict only the structured residual. This design reduces the effective learning problem from the full path-loss range to a narrower residual distribution, improving data efficiency and interpretability.

Try 78 implements this philosophy for path loss: a coherent two-ray model fitted per height bin provides a prior map $\hat{y}_{\mathrm{PL}}(\mathbf{x}) = \mathrm{TwoRay}(d, h_{\mathrm{tx}})$, and the neural component predicts the residual $\epsilon = y - \hat{y}_{\mathrm{PL}}$. Try 79 extends the idea to delay spread and angular spread by using a ridge-regression model in log space with TX-centred geometric features and local morphology statistics. Try 80 unifies all three targets under a single shared network anchored to both Try 78 and Try 79 priors.

2.5 Channel statistics prediction: delay spread and angular spread

2.5.1 Empirical models for spread parameters

Standard channel models characterise delay spread and angular spread as log-normal distributions parameterised by distance and LoS/NLoS state. TR 38.901 [1] provides polynomial fits for the log-normal mean and standard deviation as functions of 2D distance and transmitter height across UMa, UMi, and RMa scenarios. These fits are limited to the morphology class and frequency band of the underlying measurement campaigns.

Log-domain modelling is physically motivated: spread parameters are multiplicative quantities in physical space, so their logarithm is more Gaussian-like and additive-noise assumptions are better justified. Try 79 adopts this convention by working in $\log(1 + y)$ space throughout, consistent with the 3GPP empirical modelling framework.

2.5.2 Deep learning for wideband channel statistics

Work on deep learning for delay and angular spread is sparser than for path loss. Most attempts either predict spread globally per sample (a single scalar) rather than per pixel, or treat it as an ancillary output of a multitask path-loss model. Per-pixel dense prediction of delay spread and angular spread over 513×513 maps is, to the best of the author's knowledge, not addressed by published methods at the time of writing.

The main challenge is the spike-plus-exponential distribution shape of spread targets. A model trained only with mean-squared error (MSE) collapses to the conditional mean, which lies in the interior of the spike and the exponential tail and is therefore a poor representative of either regime. Try 77 addresses this with a Gaussian mixture model (GMM) distribution head, and Try 80 incorporates the Try 79 log-domain ridge prior as an anchor before training the residual neural head.

2.5.3 Distributional mismatch and the spike-plus-exponential problem

The spike-plus-exponential shape — a Dirac-like concentration near zero (the LoS-dominated case) plus a heavy exponential tail (NLoS multipath) — was confirmed empirically on the CKM dataset. Skewness exceeds 5 and excess kurtosis exceeds 40 for both delay and angular spread in several topology classes. A single Gaussian or even a symmetric loss function does not fit this family.

Mixture models are the natural solution. GMM-3 was identified as the best-fit family for every LoS/NLoS histogram in the CKM distribution analysis. The mixture component with the lowest mean captures the spike, and the remaining components capture the exponential tail. This motivates the Stage-A distribution head in Try 76 (for path loss) and Try 77 (for spreads), carried into Try 80 as the distributional prior for the residual head.

2.6 Distribution-aware and multi-target modelling

2.6.1 Heteroscedastic uncertainty

Kendall and Gal [16] established the framework for heteroscedastic neural networks in computer vision: predict both a mean output $\mu(x)$ and a log-variance $\log \sigma^2(x)$, optimising the negative log-likelihood (NLL)

$$\mathscr{L} = \frac{(\mu - y)^2}{\sigma^2} + \log \sigma^2. \tag{2.3}$$

Pixels with high aleatoric uncertainty (deep NLoS shadows, building edges) receive lower effective weight, while the network is penalised for overestimating uncertainty everywhere. Try 71 of this thesis applies this loss to path loss prediction, with σ^2 predicted as a second output channel of PMHHNet. Evaluation curves of RMSE versus uncertainty-threshold coverage showed that filtering the highest-uncertainty pixels substantially reduces overall RMSE.

2.6.2 Gaussian mixture model heads

Beyond heteroscedastic Gaussians, the GMM head allows multi-modal output distributions. Try 76 implements a two-stage architecture: Stage A produces per-sample GMM parameters $\{\pi_k, \mu_k, \sigma_k\}_{k=1}^{K}$ via a shallow convolutional encoder with sinusoidal FiLM conditioning on UAV height; Stage B is a small U-Net that produces per-pixel component assignments $p_k(i, j)$ and within-component deviations $z(i, j)$. The final prediction is

$$\hat{y}(i, j) = \sum_{k=1}^{K} p_k(i, j) \cdot \big(\mu_k + z(i, j) \cdot \sigma_k\big). \tag{2.4}$$

This formulation avoids the mode-collapse failure of MSE regression while maintaining a compact architecture. The same two-stage GMM design is used in Try 77 for delay and angular spread, with modified parameters for the spike-plus-exponential prior structure.

2.6.3 Stochastic diffusion models

Conditional diffusion models applied to radio maps [24] generate physically plausible stochastic variations in deep shadow zones rather than predicting a single blurry mean. The iterative denoising procedure can be conditioned on topology maps, transmitter locations, and height values. RadioDiff reports 0.5–2 dB improvement over deterministic regression baselines, primarily in NLoS-heavy regions where the posterior distribution over path loss is genuinely multi-modal. These models are computationally expensive at inference time and are listed as a future-work horizon for this thesis.

2.6.4 Multitask and joint prediction

Training a single shared backbone across multiple output targets can improve sample efficiency and consistency between targets. Path loss, delay spread, and angular spread are physically correlated: a deep shadow corresponds to high delay spread and potentially wide angular spread from diffracted paths. A joint model can therefore use the intermediate feature maps learned for path loss to partially explain spread patterns, reducing the effective learning problem.

Try 80 instantiates this idea explicitly: one shared encoder processes the topology map and UAV height, and three parallel decoder heads predict path-loss residual, delay-spread residual, and angular-spread residual respectively, all anchored to their corresponding physical priors. The multi-task loss is a weighted sum of per-target NLL terms, with weights tuned on the validation set.

2.7 Training stabilisation and regularisation techniques

2.7.1 Stochastic weight averaging

Stochastic weight averaging (SWA) [13] averages model weights along the training trajectory at a low, approximately constant learning rate. This finds wider optima in the loss landscape that generalise better to unseen data. The CKM city-holdout evaluation makes generalisation particularly important since validation and test cities are never seen during training. Tries 67–71 incorporate SWA starting from 60 % of the training budget.

2.7.2 Exponential moving average weights

Separate from SWA, exponential moving average (EMA) tracking of the model weights at a decay rate of 0.9975 produces a smoother checkpoint suitable for validation scoring. EMA weights are used for validation and final test inference but not for gradient updates. The EMA-smoothed validation score is also used as input to the ReduceLROnPlateau scheduler to avoid reacting to noisy single-epoch validation spikes.

2.7.3 Feature-wise linear modulation

Feature-wise linear modulation (FiLM) [35, 29] conditions the activations of a neural network on a scalar or vector input by applying learned affine transforms $\gamma(c) \cdot \mathbf{h} + \beta(c)$ to intermediate feature maps. Sinusoidal positional encoding of the UAV height scalar (using log-spaced frequencies over the full 12–478 m range) is fed into FiLM layers at multiple encoder depths. This ensures that the UAV altitude continuously modulates the activation statistics rather than being handled as a discrete bin, which is critical when the effective propagation regime changes substantially with height.

2.8 Key benchmarks and fair-comparison protocol

2.8.1 Why headline numbers cannot be compared directly

The most widespread mistake in radio-map literature reviews is comparing headline RMSE values without verifying whether the underlying tasks are equivalent. The following evaluation axes are the minimum required for a valid comparison:

1. **Split:** city-holdout vs. in-distribution random split vs. same-scene split.
2. **Transmitter height:** fixed rooftop, discrete bins, or continuous UAV altitude.
3. **Building pixels:** excluded from loss, excluded from metrics, set to zero error, or included.
4. **Target quantity:** path loss (dB), path gain (dBm), received power, delay spread, angular spread.

5. **Dynamic range and clipping:** determines the RMSE floor and ceiling independently of model quality.
6. **Inference extras:** test-time augmentation (TTA), calibration, SWA, EMA weights; must be declared for both systems compared.

This thesis uses a strict city-holdout split, continuous UAV height, ground-only metrics (building pixels excluded from loss and metrics), native 513 × 513 pixel maps, and no post-training calibration. Table 2.2 summarises how published methods compare along these axes.

Table 2.2: *Evaluation condition comparison across representative methods.*

Method	Unseen city	Cont. UAV height	≥512 px	Zero calib.	Comparable?
PMNet / ICASSP 2023 [19]	No	No	No	Yes	No
RMTransformer [21]	No	No	No	Yes	No
Gao et al. [8]	No	No	No	Yes	No
AIRMap [31]	Partial	No	Yes	No	Partial
Tarhouni et al. [33]	No	No	No	Yes	Physical ref.
PathFinder [42]	Partial	No	No	Yes	Partial
Try 80 (this thesis)	**Yes**	**Yes**	**Yes**	**Yes**	**Reference**

2.8.2 Tarhouni et al. — measured suburban path-loss prediction

Tarhouni et al. [33] study machine-learning-based path-loss prediction in the sub-6 GHz band using a suburban campus environment. The paper is valuable for this thesis because it sits closer to physical path-loss prediction than purely image-domain RadioMap3DSeer challenge scores: the reported errors are in dB and the setting is tied to a concrete environment rather than only a procedural map generator. However, it is not a strict one-to-one comparison to CKM. The setup is terrestrial/suburban, lower frequency, and not a dense 513 × 513 UAV CKM with per-pixel delay and angular-spread targets. I therefore use it as an outdoor path-loss reference for the approximate difficulty of real-environment generalisation, while keeping the final comparison qualitative.

The key lesson for Try 80 is not that the numerical RMSE should match Tarhouni's setting, but that outdoor path-loss predictors evaluated outside their training distribution tend to move back toward multi-dB errors even when the same model family looks much better

under in-distribution challenge conditions. This supports the decision to report Try 80 under a strict city-holdout protocol.

2.8.3 AIRMap — adaptive digital-twin radio maps

AIRMap [31] is an adaptive U-Net autoencoder processing 512 × 512 elevation maps for ultra-low-latency digital-twin synchronisation. It reports under 5 dB RMSE (path gain), but this figure is obtained after lightweight field calibration from 20 % of the test measurements. The zero-shot baseline is also stated as under 5 dB on Boston-only training, but the gap is plausibly explained by: (i) calibrated path gain vs. uncalibrated absolute path loss, (ii) single-city training and test vs. 66-city diversity, and (iii) fixed terrestrial Tx vs. continuous UAV height.

AIRMap is nevertheless an important citation for this thesis because it shows that elevation-only deep surrogates can replace repeated ray tracing for digital-twin style workloads. Its reported inference time of about 4 ms on an NVIDIA L40S highlights the practical reason for the whole CKM surrogate direction: the value of the model is not only lower RMSE, but the ability to produce dense radio maps fast enough for planning loops. Try 80 is less mature than AIRMap as a deployable digital-twin system, but it attacks a harder multi-target UAV setting and keeps physical priors visible instead of using only a black-box elevation-to-path-gain network.

2.8.4 Gao et al. — multi-layer segmentation with corridor weighting

Gao et al. [8] propose a ResNet-based outdoor path-loss predictor with Tx depth maps, Rx depth maps, a distance map, and a weighting map that emphasises the direct Tx–Rx corridor. The authors report that this design outperforms PPNet, RPNet, and ViT baselines by roughly 1.2–3.0 dB on ITU Challenge 2024 and ICASSP 2023-style data, while reducing FLOPs by about 60%. This is highly relevant to the CKM work because it formalises a point that appeared repeatedly in the internal tries: the physically important region is not the whole topology image uniformly, but the subset of geometry that lies on or near the propagation corridor.

Try 66 through Try 70 implement only a weaker Tx-centred radial proxy of that idea. Try 78 then solves the LoS part more cleanly with a two-ray radial prior, while Try 80 uses the calibrated prior as an input channel. A full Gao-style per-Rx corridor map remains future work because generating one for every receiver pixel at 513 × 513 resolution is substantially more expensive than using the current Tx-centred distance and obstruction features.

2.8.5 PathFinder — multi-transmitter domain adaptation

PathFinder [42] tackles distribution shift via disentangled feature encoding and mask-guided low-rank attention, reporting the lowest MSE of 0.1068 on unseen rural datasets.

The explicit domain-generalisation focus aligns with the CKM city-holdout spirit, though PathFinder's setup involves multi-transmitter inputs and a different normalisation convention. Techniques from PathFinder such as transmitter-oriented mixup augmentation are listed as medium-priority future work for this thesis.

This citation matters because it separates two issues that are often mixed in radio-map papers: prediction accuracy under the training distribution and robustness under distribution shift. PathFinder explicitly argues that many methods over-focus on single-transmitter, in-distribution settings and under-test multi-transmitter or shifted building-density regimes. Try 80 is not a multi-transmitter model, but its city-holdout split is motivated by the same concern: a CKM surrogate is only useful if it works on new urban layouts, not only on shuffled samples from the same city.

2.8.6 ICASSP 2025 indoor challenge

The ICASSP 2025 Indoor Pathloss Challenge [2] presents a structurally different problem — dense wall-penetration NLoS effects in bounded interior spaces — but its public results are a useful warning against over-interpreting sub-2 dB outdoor challenge numbers. The official result page reports that the winning SIP2Net submission obtains a weighted-average RMSE of 9.411 dB; IPP-Net reaches 9.501 dB; TerRaIn reaches 10.325 dB; and TransPathNet reaches 10.397 dB [11, 22, 3, 20]. These are not outdoor UAV results, but they show that when NLoS physics becomes harder and less directly observable from the input map, modern neural architectures again hit a roughly 9 dB to 10 dB error band.

The comparison to Try 66 should therefore be read as diagnostic rather than competitive. Try 66's 9.41 dB path-loss result is not claimed to beat indoor methods; it is used to argue that the earlier CKM direct-regression family had reached the same type of NLoS-dominated wall. Try 78 and Try 80 were developed precisely because improving that wall required a different formulation: strong priors, regime separation, and residual distribution modelling.

2.9 Emerging techniques for closing the NLoS gap

The 9–10 dB barrier common to both indoor and outdoor NLoS is not a dataset scaling artefact; it reflects the fundamental difficulty of inferring volumetric wave propagation from a 2D building height map. Closing this gap requires either richer inputs, stronger physical constraints, or stochastic modelling of the aleatoric uncertainty. Table 2.3 summarises the techniques with the highest estimated impact for CKM-style evaluation.

Items 1–3 are partially implemented in the current thesis; items 4–6 are roadmap items requiring additional data, compute, or infrastructure beyond the scope of a single-GPU training setup.

Table 2.3: *Emerging techniques ranked by expected impact on CKM NLoS RMSE.*

#	Technique	Key reference	Est. gain	Complexity
1	Knife-edge diffraction prior	[7]	1–3 dB NLoS	Medium
2	PINN Helmholtz residual	ReVeal [28]	0.5–2 dB	High
3	Dual LoS/NLoS heads	Tries 69, 71	1–2 dB	Low
4	Test-time adaptation	RadioPiT [26]	0.3–1 dB	Low–Med
5	Diffusion refinement	RadioDiff [24]	0.5–2 dB	High
6	Foundation pre-training	FM-RME [29]	0.3–1 dB	Very high

2.10 Position of Try 80 in the literature

2.10.1 What is novel about Try 80

Try 80 occupies a distinct position in the radio-map literature through the combination of three features that no published method addresses simultaneously:

- **Three simultaneous dense targets.** Try 80 jointly predicts per-pixel path loss, delay spread, and angular spread over 513×513 maps with a single shared backbone. Published radio-map work addresses either path loss or spread statistics, but not all three in a pixel-dense city-holdout setting.
- **Frozen physics anchors.** Two independently calibrated physics-based priors — the coherent two-ray model (Try 78) for path loss and the log-space ridge regression (Try 79) for both spreads — are pre-computed and frozen. The neural network predicts residuals with respect to these anchors rather than raw targets. This reduces the effective learning range, improves interpretability, and allows the analytical prior to handle the bulk of the LoS structure without requiring the neural component to re-learn it.
- **Strict city-holdout evaluation with continuous UAV height.** All 66 CKM cities appear without repetition in train, validation, and test partitions. The UAV transmitter height is a continuous scalar in $[12, 478]$ m conditioned via sinusoidal FiLM, not a discrete bin. This combination is not used in any published benchmark at the time of writing.

2.10.2 What Try 80 does not claim

Try 80 does not claim to replace full ray-tracing in arbitrary scenarios. The CKM dataset is a ray-traced simulation, and the thesis target is to reproduce its outputs as accurately as possible under a strict generalisation protocol. Physical effects that are not represented in the 2D topology inputs — such as material-dependent penetration loss, antenna radiation patterns, or atmospheric effects — are outside the prediction scope.

The thesis also does not claim parity with RadioMap3DSeer challenge numbers. As discussed in Section 2.8, those numbers apply to a structurally easier evaluation contract (fixed height, same-distribution cities, narrower dynamic range) and cannot be compared to CKM city-holdout RMSE without systematic correction of all five axes listed in that section.

2.10.3 Summary of position

Table 2.4 provides a compact summary of how Try 80 compares to the most relevant published work across the evaluation dimensions introduced in this chapter.

Table 2.4: *Positioning of Try 80 against the most relevant published methods.*

Benchmark / Method	**True City holdout RMSE (path loss)**		**Verdict**
ICASSP 2023 best [21]	0.97 dB	No	Different problem
AIRMap [31] (calibrated)	< 5 dB	Partial	Different protocol
Gao et al. [8] (multi-layer)	5.6 dB	No	Different protocol
Tarhouni et al. [33]	multi-dB, sub-6 GHz	No	Physical outdoor reference
Try 66 (this thesis, single expert)	9.41 dB	Yes	Internal baseline
Try 78 non-DL prior	≈1.75 dB LoS	Yes	LoS-only anchor
Try 80 (this thesis)	**target <5 dB**	**Yes**	**This work**

The < 5 dB path-loss target for Try 80 is framed as a project goal rather than a published result, consistent with the thesis performance targets stated in the introduction. The prior-anchored design is expected to allow the neural residual head to converge faster and to a narrower error floor than the direct-regression family of Tries 66–71, particularly in LoS-dominated morphology classes where the physics anchor already achieves sub-2 dB error.

CHAPTER 3

Methodology / project development

This chapter describes the dataset protocol, the evolution of the modelling approach, and the final Try 80 architecture. The emphasis is on decisions that affect reproducibility: masking, splitting, conditioning, and the distinction between raw deep prediction and prior-anchored residual prediction.

3.1 Dataset protocol

Each sample is a complete 513×513 pixel CKM. The model never receives a random crop as an independent training city. Instead, the split is performed at city level. For the final Try 76, Try 77, and Try 80 family the split uses `split_seed = 42`, with 15% validation and 15% test samples after city assignment. This avoids the most common optimistic error in spatial radio prediction: training on one part of a city and testing on another visually similar part of the same city.

The receiver mask is:

$$m_{\text{ground}}(i,j) = \begin{cases} 1, & \text{if topology}(i,j) = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{3.1}$$

All RMSE values in this thesis are pixel-weighted over valid ground pixels:

$$\text{RMSE} = \sqrt{\frac{\sum_{s,i,j} m_s(i,j)\,(\hat{y}_s(i,j) - y_s(i,j))^2}{\sum_{s,i,j} m_s(i,j)}}. \tag{3.2}$$

This is stricter than averaging per-sample RMSEs because large valid maps contribute proportionally to the number of receiver pixels.

3.2 From U-Net and cGAN to masked baselines

The first implementation used a U-Net generator with a PatchGAN discriminator. The generator predicted channel maps from topology-related inputs, while the discriminator

encouraged realistic local patches. This was useful as a software prototype but not as a final scientific direction. The adversarial term did not improve the physical errors that dominated validation RMSE, and it made diagnosis harder.

The next phase focused on controlled engineering changes. Transposed convolutions were replaced by bilinear upsampling and projection to reduce checkerboard artifacts. Multi-scale supervision was added so that the model could learn both global radial structure and local map details. Try 22 became the strongest clean pre-prior path-loss baseline in this family.

After Try 33, building pixels were excluded from both training loss and validation metrics. This changed the interpretation of earlier numbers: any method that implicitly benefited from easy building pixels was no longer acceptable. The ground-only rule became an invariant for later tries.

3.3 Chronological attempt analysis

The work was not a single clean jump from U-Net to the final Try 80 model. It was closer to an engineering search in which each group of attempts tested one hypothesis and either kept it, rejected it, or transformed it into a stricter version. Table 3.1 summarizes the most important steps. Values are included only when the metric contract is sufficiently clear; otherwise the row records the technical lesson rather than forcing an unreliable comparison.

This history changed the interpretation of the project. The final model is not only the largest or latest network. It is the result of four negative lessons: GANs were not reliable RMSE optimizers for this physical target; building pixels had to be removed from the metric; pure CNNs underlearned radial LoS structure; and NLoS errors were heavy-tailed enough to require regime-aware or distribution-aware treatment. These lessons justify the final prior-anchored formulation.

3.4 Physical priors and PMHHNet

Attempts 41 to 75 moved from direct map prediction to residual learning, as formulated in Eq. 3.3. This phase was heavily influenced by the `PMNet` architecture [19], which established the viability of using U-Net-like backbones for path-loss map prediction in terrestrial cellular networks.

$$\hat{\mathbf{Y}} = \mathbf{P}_{\text{phys}} + f_\theta\big(\mathscr{T}, h, \mathbf{P}_{\text{phys}}\big), \tag{3.3}$$

where $\mathbf{P}_{\text{phys}}$ is the analytical or calibrated prior map and f_θ predicts only the residual correction that the prior cannot explain.

3.4.1 Mathematical Evolution: From PMNet to PMHHNet

The `PMHHNet` (Path-loss Map with High-frequency and Height-awareness Network) was developed as a specific refinement of the `PMNet` philosophy tailored for the UAV-enabled Air-to-Ground (A2G) scenario.

High-Frequency (HF) Stem. Standard convolutional encoders tend to smooth out sharp building transitions during downsampling. To counter this, `PMHHNet` introduces a dedicated high-frequency stem. Let $\mathscr{T}$ be the topology map. We compute the discrete Laplacian $|\nabla^2\mathscr{T}|$ to identify building edges and shadow boundaries. The initial feature map $\mathbf{X}_0$ is computed as:

$$\mathbf{X}_0 = \text{Stem}(\mathscr{T}) + \text{Proj}(|\nabla^2\mathscr{T}|) \tag{3.4}$$

where Proj is a 1×1 convolution that maps the edge-detector output to the stem's feature depth. This ensures that building geometry is strongly injected into the earliest layers of the encoder.

Continuous Sinusoidal FiLM Conditioning. While `PMNet` was designed for fixed base-station heights, UAVs require continuous height generalization. `PMHHNet` utilizes Feature-wise Linear Modulation (FiLM) [23] to inject the scalar height h. The height is first mapped to a sinusoidal embedding $\mathbf{e}(h) \in \mathbb{R}^K$ (Eq. 3.5). For any feature map $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ at depth l, the conditioned map $\mathbf{X}'$ is:

$$\mathbf{e}(h) = \left[\sin(\omega_1\tilde{h}), \cos(\omega_1\tilde{h}), \dots, \sin(\omega_{K/2}\tilde{h}), \cos(\omega_{K/2}\tilde{h})\right], \qquad \tilde{h} = \frac{h - h_{\min}}{h_{\max} - h_{\min}}, \tag{3.5}$$

$$\mathbf{X}' = \mathbf{X} \odot (1 + \mathbf{g}_l(\mathbf{e}(h))) + \mathbf{b}_l(\mathbf{e}(h)) \tag{3.6}$$

where $\mathbf{g}_l$ and $\mathbf{b}_l$ are learned linear projections mapping the embedding to the channel dimension C. This pattern, inspired by time-step conditioning in diffusion models [6], allows the network to modulate its internal representation dynamically based on the UAV's altitude.

3.4.2 Limitations and Legacy

Despite these improvements, the early residual phase plateaued at about 9.4 dB (Try 66). The primary bottleneck was the "soft" nature of the residual learning; the network often under-compensated for the physical propagation law in NLoS regions.

> *Legacy of PMHHNet:* While the full architecture was eventually superseded by the prior-anchored GMM approach of `Try 80`, its core components—the HF stem and sinusoidal FiLM—remain highly effective for point-prediction tasks or real-time implementations where the distributional complexity of a GMM is not required.

3.5 Distribution-first modelling

The decisive diagnosis after Try 75 was that NLoS path loss was not merely a harder regression target; it possessed a fundamentally different error distribution. Try 76 addressed this by predicting the parameters of a Gaussian Mixture Model (GMM) rather than a simple point estimate.

Try 77 applied the same idea to delay spread and angular spread. These targets are closer to a "spike-plus-tail" distribution (a spike near zero for LoS, and a heavy tail for NLoS).

For path loss, Try 76 utilized 12 specialized experts: six topology classes partitioned by LoS and NLoS regions. As shown in Figure 3.1, Stage A acts as a distribution head, predicting the global GMM parameters (π_k, μ_k, σ_k) from map features and altitude conditioning. Stage B then decodes these priors spatially, predicting per-pixel soft component assignments and within-component offsets. The final map reconstruction can be written as:

$$\hat{y}(i,j) = \sum_{k=1}^{K} p_k(i,j)\,(\mu_k + z_k(i,j)\sigma_k)\,. \tag{3.7}$$

Try 77 extended this distribution-first philosophy to delay spread and angular spread. As illustrated in Figure 3.2, these targets are not cleanly Gaussian; their empirical distributions resemble a near-zero spike (representing the LoS direct path) combined with a heavy tail (representing NLoS scattering). Standard Mean Squared Error (MSE) objectives tend to underrepresent these rare but physically significant heavy-tail spread values, necessitating a dedicated mixture head.

3.6 Try 78 and Try 79 priors: overview and literature basis

This section provides the technical documentation for the non-DL physical priors (`Try 78` and `Try 79`) and the final joint architecture (`Try 80`). It serves as a bridge between the high-level attempt history and the numerical results, explaining the exact mathematical transformations, how every parameter was fitted in practice, and what is genuinely new in the final `Try 80` approach.

Notation convention. Following the notation used in the ETSETB template, vectors and matrices are typeset in bold roman font. Vectors use lowercase symbols and matrices, maps or tensors use uppercase symbols. Scalars are left in italic math font. For example, $\mathbf{x}_n$ denotes the feature vector of pixel n, $\mathbf{X}$ denotes a design matrix, $\mathbf{w}$ denotes a vector of fitted coefficients, and y_n, $d_{2D,n}$, or h_{tx} denote scalar quantities. The implementation often stores full maps as tensors, but the derivations below use a pixel index n whenever this makes the regression or likelihood clearer.

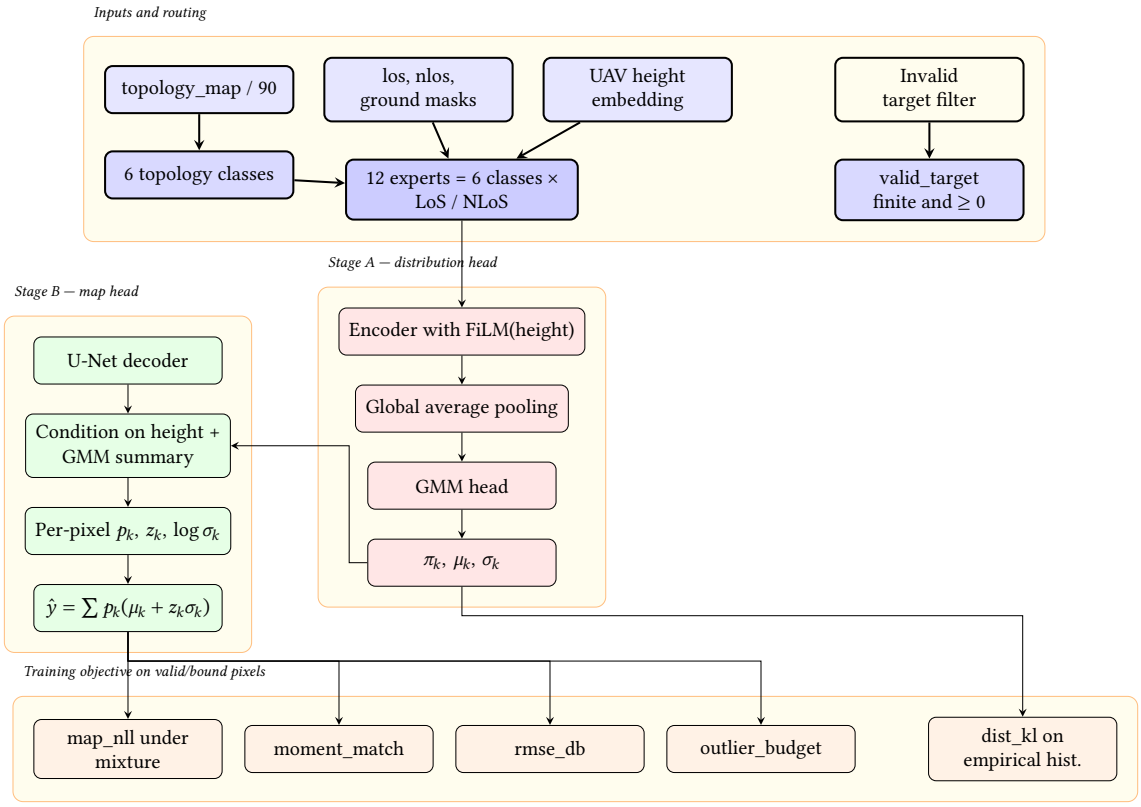

Figure 3.1: *Try 76 distribution-first path-loss system. Stage A learns global GMM parameters (π_k, μ_k, σ_k) from pooled features conditioned on UAV height; Stage B produces per-pixel soft assignments and within-component offsets anchored to those parameters.*

Block	Literature basis	CKM adaptation	Role in `Try 80`
Path-loss LoS	FSPL and coherent two-ray propagation	Effective height-dependent reflection parameters and radial residuals fitted from CKM LoS pixels	Frozen LoS prior, injected as an input channel and used as the anchor for bounded residual learning.
Path-loss NLoS	COST-231/Hata-style urban loss and A2G elevation envelopes	Regime-wise calibration with local density, height, shadowing and elevation features	Frozen NLoS prior selected by the explicit LoS/NLoS geometry mask.
Delay and angular spread	Log-domain large-scale spread modelling and LoS/NLoS separation	Regime-wise ridge regression in $\log(1 + y)$ space using multiscale morphology and obstruction features	Frozen spread priors used as anchors for the multi-task residual model.
`Try 80` residual model	Probabilistic regression and mixture-density ideas	Prior-anchored per-pixel GMM with residual caps, anchor gates, explicit region masks and shared multi-task trunk	Learns residual uncertainty and local corrections without replacing the interpretable priors.

Table 3.2: *One-page summary of what is inherited from the literature, what is adapted to CKM, and what is reused or newly assembled in `Try 80`.*

3.6.1 High-level architecture

The central design decision in `Try 80` is *not* to learn path loss, delay spread, and angular spread from scratch. Instead:

- `Try 78` provides a frozen prior for `path_loss`,
- `Try 79` provides frozen priors for `delay_spread` and `angular_spread`,
- `Try 80` learns bounded residual corrections around those priors.

The deterministic or interpretable structure is kept in the prior, and the deep model is used only for the residual part that the prior cannot explain. Figure 3.3 illustrates the overall pipeline.

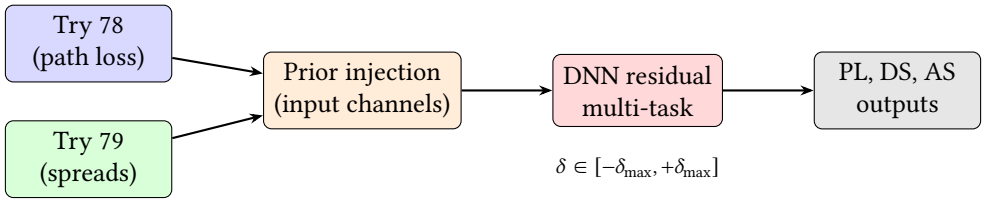

Figure 3.3: *Overall* `Try 80` *pipeline. The frozen priors are injected into the DNN, which learns a bounded correction δ around them.*

3.6.2 What was reused from the literature, what was changed, and what is new

The `Try 80` approach is not a single literature formula, but a multi-stage construction. The following points summarize the technical decisions for each task.

Try 78: path loss

What the literature provides. FSPL is a textbook formula [27] that gives the power loss of a plane wave in free space as a function of distance and frequency. The two-ray model for a transmitter above a flat reflecting ground is also classical [27]. The COST-231 extension of the Hata model is a standard empirical formula for macrocell path loss in the 1.5–2 GHz range. Treating LoS and NLoS as fundamentally different propagation regimes is standard practice in 3GPP TR 38.901 [1].

What we took exactly from the literature. The FSPL formula is used verbatim. The two-ray concept—direct ray plus ground-reflected ray with a complex-valued sum—is taken directly from the standard derivation in [27]. The COST-231 formula is used verbatim including all empirical constants. The qualitative idea of LoS/NLoS regime separation is taken from [1].

What we changed or did not take.

- **Reflection coefficient Γ.** Instead of material-based Fresnel coefficients, Γ is replaced by an *effective* complex parameter $\rho(h_{\text{tx}})e^{j\phi(h_{\text{tx}})}$ fitted per UAV height bin from data.
- **Height-bin fitting.** Independent (ρ, ϕ, b) triples are fitted per 5 m height bin, producing a height-dependent effective model necessary because the CKM dataset covers roughly 10–460 m of UAV height.
- **Radial residual correction.** A separate range-dependent correction term $r(d_{2D})$ is added to capture systematic smooth trends in the residuals of the two-ray fit.
- **Regime-wise linear calibration on top of COST-231.** A 15-feature design vector $\mathbf{x}_{78}(x)$ and regime-wise OLS calibration correct the raw prior using local

density, height, NLoS support, and elevation features. This step has no direct literature equivalent.

Try 79: delay spread and angular spread

What the literature provides. 3GPP TR 38.901 [1] and WINNER II [18] model delay spread and angular spread as log-normal random variables. Separate parameters are given for LoS and NLoS, and for different scenario types. Empirical UAV studies [4, 14] report that delay and angular spread increase as the UAV elevation angle decreases, which was corroborated by inspecting the dataset data.

What we changed or did not take.

- **Log-domain modelling.** We reuse the literature assumption that spreads are log-normally distributed, so we predict $\log(1 + y)$.
- **Morphology features.** Standard models use distance and antenna height; we extend this with local building density and height features extracted from the CKM map.

Try 80: architecture and loss

- **No stochastic channel simulator.** We fit a deterministic regression that predicts the *expected* spread value at each map pixel, conditioned on local geometry.
- **Map-derived morphology features.** Per-pixel features such as local building density $\rho_k(x)$, mean building height $h_k(x)$, NLoS-support fraction $n_k(x)$, blocker depth $b_{41}(x)$, transmitter clearance $c_{41}(x)$, and taller-building fraction $t_{41}(x)$ have no equivalent in 3GPP or WINNER.
- **Regime-wise ridge regression per pixel.** The regime-wise ridge regression and the associated fallback hierarchy are entirely our construction, designed to handle the sparse-regime problem that arises when the dataset is stratified across metric × topology × LoS/NLoS × antenna_bin.
- **Shared UNet with FiLM.** The backbone is a standard medical/vision U-Net, but with sinusoidal FiLM conditioning on UAV height.
- **Prior-anchoring.** This is a genuinely new design choice for this project: the model can only learn small corrections $[-\delta, +\delta]$ relative to the physical priors.
- **UAV height as a feature.** The normalized height h_{norm} and its square h_{norm}^2 are included because the CKM dataset spans a wide range of UAV heights.

3.7 Try 78: Path Loss Prior

3.7.1 LoS branch: coherent two-ray model

Geometry and distances

Consider a UAV transmitter at height h_{tx} (metres above the ground-plane reference) and a receiver at height h_{rx}. For each map pixel x at 2D horizontal distance $d_{2D}(x)$ from the transmitter, two propagation distances are defined.

The *direct-path* (LoS) distance is:

$$d_{\text{LoS}}(x) = \sqrt{d_{2D}(x)^2 + (h_{\text{tx}} - h_{\text{rx}})^2} \tag{3.8}$$

This is simply the Euclidean distance between the two antennas in 3D space. Because both $d_{2D}(x)$ and the height difference $(h_{\text{tx}} - h_{\text{rx}})$ are non-negative reals, $d_{\text{LoS}}(x) \geq d_{2D}(x)$, with equality only when both antennas are at the same height.

The *reflected-path* (image) distance is:

$$d_{\text{ref}}(x) = \sqrt{d_{2D}(x)^2 + (h_{\text{tx}} + h_{\text{rx}})^2} \tag{3.9}$$

This is the distance from the image source (the UAV mirrored below the ground plane at $-h_{\text{tx}}$) to the receiver. Since $(h_{\text{tx}} + h_{\text{rx}}) \geq |h_{\text{tx}} - h_{\text{rx}}|$ for any non-negative heights, we always have $d_{\text{ref}}(x) \geq d_{\text{LoS}}(x)$, reflecting the fact that the reflected path is always longer than the direct path.

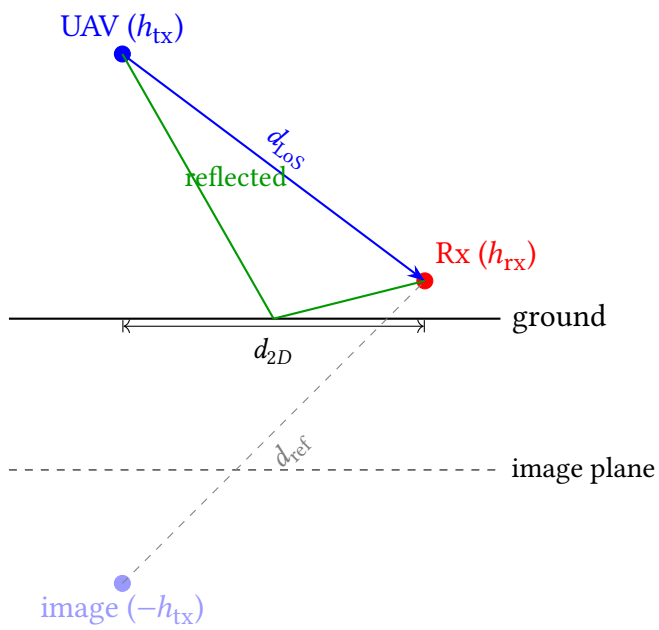

Figure 3.4: *Two-ray geometry. Direct path (blue) has length d_{LoS}; reflected path (green) has the same total length as the image-source path (gray dashed), which is d_{ref}.*

Free-space path loss

The free-space path loss in dB is defined using the standard formula at frequency f_{MHz} (in MHz):

$$\mathrm{FSPL}(x) = 32.45 + 20\log_{10}\left(\frac{d_{LoS}(x)}{1000}\right) + 20\log_{10}(f_{MHz}) \tag{3.10}$$

Derivation. FSPL in linear power ratio is $(4\pi d/\lambda)^2$ where $\lambda = c/f$. In dB:

$$\begin{aligned}
\mathrm{FSPL}_{dB} &= 20\log_{10}\left(\frac{4\pi d f}{c}\right) \\
&= 20\log_{10}(4\pi) + 20\log_{10}(d) + 20\log_{10}(f) - 20\log_{10}(c) \\
&\approx 20\log_{10}(d_{km}\cdot 1000) + 20\log_{10}(f_{MHz}\cdot 10^6) + 20\log_{10}(4\pi) - 20\log_{10}(3\times 10^8) \\
&= 20\log_{10}(d_{km}) + 20\log_{10}(f_{MHz}) + 32.45 \quad [\mathrm{dB}]
\end{aligned} \tag{3.11}$$

The constant $32.45 = 20\log_{10}(4\pi\cdot 10^3\cdot 10^6/(3\times 10^8)) \approx 32.45$ dB. In Eq. (3.10), $d_{LoS}(x)/1000$ converts metres to kilometres.

Physical meaning. FSPL grows quadratically with distance ($\propto d^2$) and quadratically with frequency. It captures only the geometric spreading of the wavefront, with no ground interaction, scattering, or diffraction.

Coherent two-ray correction

In the two-ray model, the received field is the superposition of the direct path and a ground-reflected path. Let $k = 2\pi f/c$ be the free-space wavenumber. The complex sum of the two waves at the receiver is:

$$E_{\text{total}} \propto \frac{e^{-jkd_{\text{LoS}}}}{d_{\text{LoS}}} + \Gamma \frac{e^{-jkd_{\text{ref}}}}{d_{\text{ref}}} \tag{3.12}$$

where Γ is the complex ground reflection coefficient. In the standard textbook derivation, Γ is computed from the Fresnel equations for the ground material. Here, it is replaced by an *effective* complex parameter:

$$\Gamma(h_{\text{tx}}) = \rho(h_{\text{tx}})\, e^{j\phi(h_{\text{tx}})} \tag{3.13}$$

where $\rho \in [0, 1.5]$ is the effective amplitude (allowing slight over-unity to capture constructive multipath effects) and $\phi \in [-\pi, \pi]$ is the effective phase offset. Both depend on the UAV height bin.

The two-ray path-loss correction term in dB is:

$$C_{\text{2ray}}(x) = -20 \log_{10} \left| 1 + \rho(h_{\text{tx}}) \frac{d_{\text{LoS}}(x)}{d_{\text{ref}}(x)} e^{-j\left(k\left(d_{\text{ref}}(x) - d_{\text{LoS}}(x)\right) + \phi(h_{\text{tx}})\right)} \right| \tag{3.14}$$

Term-by-term interpretation.

- $\rho(h_{\text{tx}})$: amplitude of the ground reflection. A value of $\rho = 0$ recovers pure FSPL. The fitted values are $\rho \approx 0.49$–0.56 for heights 10–100 m, consistent with a moderate ground-reflection coefficient.
- $d_{\text{LoS}}(x)/d_{\text{ref}}(x)$: path amplitude ratio. This is always ≤ 1 since the reflected path is longer; it corrects for the $1/d$ amplitude decay of the reflected ray relative to the direct ray.
- $k(d_{\text{ref}}(x) - d_{\text{LoS}}(x))$: the phase difference accumulated due to the extra path length of the reflected wave.
- $\phi(h_{\text{tx}})$: an additional phase offset capturing phase shifts upon reflection (e.g., phase shift at the boundary). The fitted values are $\phi \approx 0$ for all height bins, meaning in-phase reflection dominates.
- $-20 \log_{10} |\cdot|$: the total field amplitude at the receiver, normalized to the direct-path field. When the two waves add in phase, the field amplitude is > 1, giving a *negative* correction (the signal is *stronger* than FSPL alone); when they cancel, the correction is *positive* (fading loss).

This spatial interference pattern creates the characteristic ring-like behavior in the LoS path-loss maps: concentric rings of constructive and destructive interference around the transmitter.

Comparison with literature two-ray models

The coherent two-ray model is a staple of wireless propagation theory, but our implementation in `Try 78` deviates from the standard textbook treatment in several ways.

Physical vs. effective parameters. In classic texts such as Rappaport [27] and Goldsmith [9], the complex reflection coefficient Γ is derived from the Fresnel equations for a smooth dielectric surface:

$$\Gamma_{\perp} = \frac{\sin\theta - \sqrt{\epsilon_c - \cos^2\theta}}{\sin\theta + \sqrt{\epsilon_c - \cos^2\theta}}, \quad \Gamma_{\parallel} = \frac{\epsilon_c \sin\theta - \sqrt{\epsilon_c - \cos^2\theta}}{\epsilon_c \sin\theta + \sqrt{\epsilon_c - \cos^2\theta}} \tag{3.15}$$

where θ is the grazing angle and $\epsilon_c = \epsilon_r - j\sigma/\omega\epsilon_0$ is the complex permittivity of the ground. This requires assuming physical constants for the ground (typically $\epsilon_r \approx 15$ for urban soil).

In contrast, our model (Eq. 3.12) replaces the angle-dependent $\Gamma(\theta)$ with a height-dependent effective coefficient $\Gamma(h_{\text{tx}}) = \rho e^{j\phi}$. We do not assume a flat dielectric plane; instead, ρ and ϕ are fitted to the CKM dataset. This allows the model to capture aggregate scattering effects from the real, non-flat urban terrain that a pure Fresnel calculation would miss.

The d^4 approximation. Many literature sources [27, 9] simplify the two-ray model for large distances ($d \gg h_{\text{tx}}, h_{\text{rx}}$) into the classic fourth-power law:

$$P_r(d) \approx P_t G_t G_r \frac{h_{\text{tx}}^2 h_{\text{rx}}^2}{d^4} \tag{3.16}$$

This approximation assumes a perfect reflection ($\Gamma = -1$) and small grazing angles. Our model uses the *exact* field sum (Eq. 3.12) without the d^4 approximation, which is necessary because UAV heights are often significant relative to horizontal distances, making the small-angle approximation invalid.

Over-unity amplitude ($\rho > 1$). Standard physics dictates $|\Gamma| \leq 1$. However, our calibration grid search (Sec. 3.7.2) allows $\rho \in [0, 1.5]$. While physically impossible for a single specular reflection, an effective $\rho > 1$ allows the linear basis of the two-ray model to capture constructive multipath interference from multiple scatterers (e.g., building faces near the transmitter) that align with the geometric phase of the ground ray.

Radial correction. Standard models [1] treat the two-ray or FSPL result as the final mean. Our model adds a calibrated radial residual $r(d_{2D}, h_{\text{tx}})$ (Eq. 3.17) which acts as a data-driven "mopping up" term for systematic geometric errors that neither FSPL nor the two-ray model can fully represent.

Calibrated LoS prior

The full calibrated LoS prior is:

$$\mathrm{PL}_{\mathrm{LoS}}(x) = \mathrm{FSPL}(x) + C_{\mathrm{2ray}}(x) + b(h_{\mathrm{tx}}) + r(d_{2D}(x), h_{\mathrm{tx}}) \tag{3.17}$$

where:

- $b(h_{\mathrm{tx}})$ is a height-dependent bias (dB): a constant correction that shifts the overall two-ray prediction for a given UAV height.
- $r(d_{2D}(x), h_{\mathrm{tx}})$ is a radial residual correction: a smooth function of horizontal distance (and height bin) that captures systematic over-/under-prediction of the two-ray model as a function of range. It is obtained by smoothing the empirical mean residual in each distance bin after the two-ray fit.

The four terms in Eq. (3.17) are applied sequentially: FSPL gives the geometric baseline; the two-ray correction adds the interference pattern; the bias corrects for a systematic global offset per height; and the radial term corrects for range-dependent systematic errors.

3.7.2 Calibration of the LoS two-ray model

Data preparation

All training samples are used. For each sample, valid LoS pixels are selected as those satisfying:

1. topology label = 0 (ground pixel),
2. $m_{\mathrm{LoS}}(x) > 0$ (marked as LoS),
3. the measured path loss is finite and ≥ 20 dB.

If a sample has more than 2 500 valid LoS pixels, a random subsample of 2 500 pixels is drawn (with a fixed seed). Samples are then grouped by UAV height into height bins of 5 m width:

$$h_{\mathrm{bin}} = 5 \cdot \left\lfloor \frac{h_{\mathrm{tx}}}{5} \right\rfloor \tag{3.18}$$

Each bin is further subsampled to a maximum of 8 000 pixels total.

Grid search for (ρ, ϕ)

For each height bin, (ρ, ϕ) are found by grid search:

$$\phi_{\text{grid}} = \left\{ -\pi + \frac{2\pi k}{N_\phi} \;:\; k = 0, 1, \ldots, N_\phi - 1 \right\}, \quad N_\phi = 48 \tag{3.19}$$

$$\rho_{\text{grid}} = \{0.00,\ 0.05,\ 0.10,\ \ldots,\ 1.50\} \tag{3.20}$$

For each pair $(\hat{\rho}, \hat{\phi}) \in \rho_{\text{grid}} \times \phi_{\text{grid}}$, and for each pixel x_i in the height bin, define the two-ray prediction without bias:

$$\hat{p}_i(\hat{\rho}, \hat{\phi}) = \text{FSPL}(x_i) - 20 \log_{10} \left| 1 + \hat{\rho}\, \frac{d_{\text{LoS}}(x_i)}{d_{\text{ref}}(x_i)} e^{-j\left(k(d_{\text{ref}} - d_{\text{LoS}}) + \hat{\phi}\right)} \right| \tag{3.21}$$

The optimal bias for each $(\hat{\rho}, \hat{\phi})$ pair is solved analytically as the empirical mean of residuals:

$$\hat{b}(\hat{\rho}, \hat{\phi}) = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{p}_i(\hat{\rho}, \hat{\phi}) \right) \tag{3.22}$$

where y_i is the measured path loss at pixel x_i and N is the number of pixels in the bin. The biased prediction is $\hat{p}_i + \hat{b}$, and the RMSE for the pair is:

$$\text{RMSE}(\hat{\rho}, \hat{\phi}) = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{p}_i(\hat{\rho}, \hat{\phi}) - \hat{b}(\hat{\rho}, \hat{\phi}) \right)^2} \tag{3.23}$$

The best pair is:

$$(\rho^*, \phi^*) = \arg\min_{(\hat{\rho}, \hat{\phi})} \text{RMSE}(\hat{\rho}, \hat{\phi}) \tag{3.24}$$

Post-fit Gaussian smoothing across height bins

After fitting all bins independently, a Gaussian-weighted smooth is applied to stabilize sparse high-altitude bins. Let $\{h_k\}$ be the fitted height bin centers, $\{(\rho_k^*, b_k^*)\}$ the per-bin fitted values, and $\{N_k\}$ the pixel counts used in fitting. The smoothed values are:

$$\tilde{\rho}(h) = \frac{\sum_k w_k(h)\, N_k\, \rho_k^*}{\sum_k w_k(h)\, N_k}, \qquad w_k(h) = \exp\left(-\frac{(h - h_k)^2}{2\sigma_h^2} \right), \quad \sigma_h = 10 \text{ m} \tag{3.25}$$

and similarly for b. The phase ϕ is always replaced with its smoothed version (treating it as effectively constant across heights).

Bins where both ρ_k^* and b_k^* differ noticeably from the smoothed global trend are flagged as suspicious (typically high-altitude bins with few samples), and their raw fit values are replaced by the smoothed values.

Radial residual correction

After fitting the two-ray model, a residual correction is computed:

$$r(d_{2D}) = \text{GaussSmooth}_{\sigma_r}\left(\text{clip}\left(\mu_\Delta(d_{2D}),\ -2.5,\ +2.5\right)\right) \tag{3.26}$$

where $\mu_\Delta(d_{2D})$ is the empirical mean of $(y - \hat{y}_{2\text{ray}})$ averaged over all pixels at horizontal distance d_{2D}, $\sigma_r = 1.5$ pixels is a Gaussian smoothing radius, and clipping limits the correction to ± 2.5 dB to prevent over-fitting on noisy radius bins.

Fitted parameter values

Table 3.3 shows representative fitted LoS two-ray parameters across height bins.

Table 3.3: *Representative fitted LoS two-ray parameters by height bin.*

h_{tx} (m)	ρ^*	ϕ^* (rad)	b^* (dB)
10	0.5605	0.000	−0.037
20	0.5242	0.000	−0.042
30	0.4950	0.000	−0.046
50	0.5468	0.000	−0.014
80	0.5482	0.000	−0.027
100	0.5463	0.000	−0.017
140	0.5414	0.000	+0.255
460	0.8966	−0.047	+1.521

The dominant pattern is $\rho \approx 0.49$–0.56 with $\phi \approx 0$ for heights 10–100 m, meaning a moderate-amplitude in-phase ground reflection. Biases are near-zero (< 0.05 dB) for these bins. At very high altitudes (> 140 m), the increasing positive bias and higher ρ values reflect sparser training data; those bins are stabilized by Gaussian smoothing.

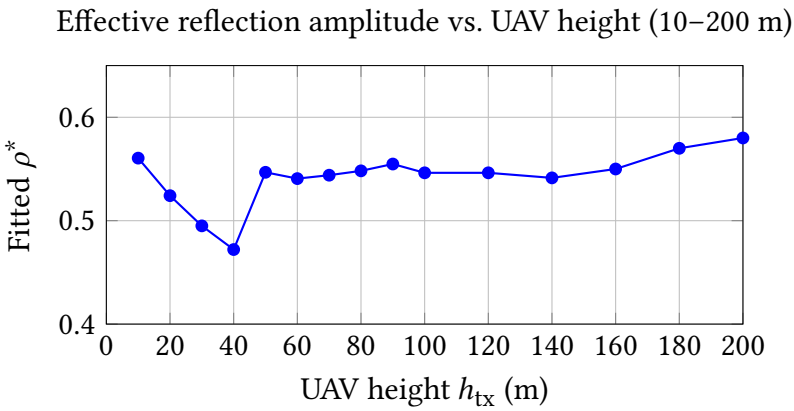

Figure 3.5: *Effective ground-reflection amplitude ρ^* as a function of UAV height, fitted independently per 5 m height bin. Values are stable in the range 0.47–0.56 for 10–130 m, then rise slightly at higher altitudes where training data is sparser.*

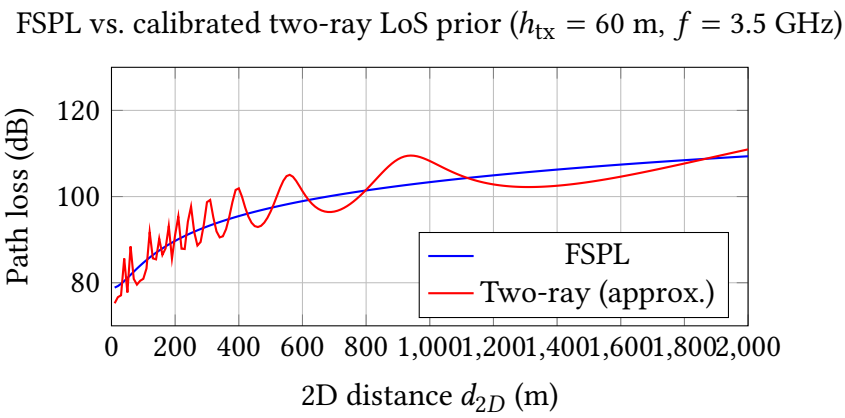

Figure 3.6: *Schematic comparison of FSPL and the coherent two-ray model for a 60 m UAV at 3.5 GHz (h_{rx} = 1.5 m). The two-ray model produces a characteristic oscillatory pattern around the FSPL baseline, corresponding to alternating zones of constructive and destructive interference.*

3.7.3 NLoS branch

The NLoS path-loss branch used in `Try 80` is not a pure literature formula. It is a hybrid engineering model built in four stages.

Stage 1: raw NLoS propagation prior

Two coarse physical estimates are combined.

COST-231 term. The COST-231 / Hata extension for urban macro cells gives the empirical large-scale path-loss expression used below [5]. It is included because it is one

of the standard references for urban NLoS macrocell loss, even though the CKM scenario is outside its original frequency and base-station height range:

$$\begin{aligned}\mathrm{PL}_{\mathrm{COST231}}(x) = 46.3 + 33.9\log_{10}(f_{\mathrm{MHz}}) - 13.82\log_{10}(h_{\mathrm{tx}}) - a(h_{\mathrm{rx}}) \\ + \big(44.9 - 6.55\log_{10}(h_{\mathrm{tx}})\big)\log_{10}(d_{\mathrm{km}}) + 3\end{aligned} \tag{3.27}$$

where $d_{\mathrm{km}} = d_{2D}(x)/1000$ and the mobile antenna correction factor for large cities is:

$$a(h_{\mathrm{rx}}) = 3.2(\log_{10}(11.75\,h_{\mathrm{rx}}))^2 - 4.97 \tag{3.28}$$

Equation (3.27) should therefore be read as a *shape prior*, not as a trusted absolute predictor. COST-231 contributes the expected log-distance growth, transmitter-height dependence and receiver-height correction of an urban empirical model. The subsequent calibration stage is what adapts that shape to the 7.125 GHz UAV ray-tracing setting used here.

Interpretation of Eq. (3.27).

- $46.3 + 33.9\log_{10}(f_{\mathrm{MHz}})$: frequency-dependent offset. The exponent 33.9 (compare to 20 for FSPL) reflects stronger frequency dependence in urban scattering.
- $-13.82\log_{10}(h_{\mathrm{tx}})$: as the base station / UAV gets higher, path loss decreases (better clearance above rooftops). The -13.82 coefficient is a fitted constant from the Hata empirical dataset.
- $-a(h_{\mathrm{rx}})$: accounts for the receiver height. For typical pedestrian heights ($h_{\mathrm{rx}} \approx 1.5$ m), $a \approx -0.6$ dB, which is small.
- $(44.9 - 6.55\log_{10}(h_{\mathrm{tx}}))\log_{10}(d_{\mathrm{km}})$: distance-dependent term. The effective path-loss exponent is $(44.9 - 6.55\log_{10}(h_{\mathrm{tx}}))/10$, which decreases as the UAV climbs (effectively reducing NLoS scattering losses at longer distances from higher altitudes).
- $+3$: the COST-231 metropolitan correction constant (dB), corresponding to the dense-urban correction in the original model [5].

Elevation-aware UAV envelope. The A2G NLoS envelope models the elevation-angle dependence of urban air-to-ground path loss. The dependence on elevation angle follows the common UAV channel-model observation that links at larger elevation angles have fewer obstructions and lower shadowing variance [17, 1].

$$\mathrm{PL}_{\mathrm{A2G,NLoS}}(x) = \lambda_0(h_{\mathrm{tx}}, f) + \beta_{\mathrm{NLoS}} + A_{\mathrm{NLoS}}\exp\left(-\frac{90 - \theta^\circ(x)}{\tau_{\mathrm{NLoS}}}\right) \tag{3.29}$$

where:

$$\lambda_0(h_{\mathrm{tx}}, f) = 20\log_{10}\left(\frac{4\pi h_{\mathrm{tx}} f}{c}\right) \tag{3.30}$$

The fitted constants are: $\beta_{\text{NLoS}} = -16.16$ dB, $A_{\text{NLoS}} = 12.0436$, $\tau_{\text{NLoS}} = 7.52$ deg.

Interpretation of Eq. (3.29).

- λ_0: a free-space-like offset that scales with UAV height and frequency [17, 32]. The CKM dataset uses $f = 7.125$ GHz (FR2 lower edge per 3GPP NR [1]); for $h_{\text{tx}} = 50$ m at that frequency, $\lambda_0 \approx 103$ dB.
- $\beta_{\text{NLoS}} = -16.16$ dB: a baseline NLoS attenuation offset relative to λ_0.
- $A_{\text{NLoS}} \exp(-\ldots)$: an exponential modulation with $(90 - \theta^\circ)$. When the elevation angle θ° is high (UAV nearly overhead, $\theta^\circ \to 90^\circ$), the exponent vanishes and the term goes to A_{NLoS}. When θ° is low (grazing incidence), the exponent is large and the term approaches zero. The effect is that NLoS path loss *decreases* as the UAV moves overhead, consistent with improved clearance above buildings.
- $\tau_{\text{NLoS}} = 7.52^\circ$: the characteristic angular scale. A low value means the transition from grazing to overhead happens sharply within a small elevation range.

The raw NLoS prior is then the conservative (higher) of the two estimates:

$$\text{PL}_{\text{NLoS,raw}}(x) = \max\left(\text{PL}_{\text{COST231}}(x), \text{PL}_{\text{A2G,NLoS}}(x)\right) \tag{3.31}$$

The maximum in Eq. (3.31) is deliberately pessimistic. If the empirical COST-231 curve predicts stronger attenuation than the A2G envelope, the model keeps the urban-macro value; if the elevation-aware envelope predicts stronger attenuation, the model keeps the A2G value. This prevents the raw prior from becoming unrealistically optimistic in NLoS streets where neither source model is fully valid on its own.

The full raw prior blends the LoS and NLoS estimates using the LoS mask $m_{\text{LoS}}(x) \in [0, 1]$:

$$\text{PL}^{(0)}(x) = m_{\text{LoS}}(x)\,\text{PL}_{\text{LoS,blend}}(x) + (1 - m_{\text{LoS}}(x))\,\text{PL}_{\text{NLoS,raw}}(x) \tag{3.32}$$

with a soft blend for the LoS branch:

$$\text{PL}_{\text{LoS,blend}}(x) = 0.7\,\text{PL}_{\text{LoS,path}}(x) + 0.3\min\left(\text{PL}_{\text{LoS,path}}(x), \text{PL}_{\text{A2G,LoS}}(x)\right) \tag{3.33}$$

The 0.7/0.3 blend in Eq. (3.33) gives 70% weight to the two-ray LoS model and 30% weight to the minimum of the two-ray model and an A2G LoS envelope. This prevents the prediction from exceeding a physically motivated upper bound in the LoS case.

Stage 2: map-derived morphology features

From $\text{PL}^{(0)}(x)$ and the map topology, a 15-element feature vector is computed:

$$\mathbf{x}_{78}(x) = \begin{bmatrix} \mathrm{PL}^{(0)}(x)^2 \\ \mathrm{PL}^{(0)}(x) \\ \log(1 + d_{2D}(x)) \\ \rho_{15}(x) \\ \rho_{41}(x) \\ h_{15}(x) \\ h_{41}(x) \\ \rho_{41}(x)\log(1 + d_{2D}(x)) \\ n_{15}(x) \\ n_{41}(x) \\ n_{41}(x)\log(1 + d_{2D}(x)) \\ \sigma_{\text{shadow}}(x) \\ \theta_{\text{norm}}(x) \\ n_{41}(x)\theta_{\text{norm}}(x) \\ 1 \end{bmatrix} \tag{3.34}$$

The lowercase bold notation is intentional: $\mathbf{x}_{78}(x)$ is a per-pixel feature vector. When vectors from many pixels in one regime are stacked for calibration, they form the uppercase design matrix $\mathbf{X}_r$. Table 3.4 gives the role of each entry before the detailed definitions below.

Term	Mathematical role	Physical interpretation
$\mathrm{PL}^{(0)}(x)^2$	Quadratic basis term.	Lets the linear calibration bend the raw path-loss curve when the raw model is systematically curved.
$\mathrm{PL}^{(0)}(x)$	Linear raw-prior term.	Keeps the calibrated model anchored to the physical prior instead of relearning path loss from scratch.
$\log(1 + d_{2D}(x))$	Smooth distance basis.	Mirrors log-distance propagation laws used in classical models [27, 5].
$\rho_{15}(x), \rho_{41}(x)$	Local building-density averages.	Count how much nearby urban structure can block, reflect or diffract the signal.
$h_{15}(x), h_{41}(x)$	Density-weighted local height averages.	Distinguish low dense blocks from taller blocks with stronger obstruction.
$\rho_{41}(x)\log(1 + d_{2D}(x))$	Density-distance interaction.	Allows distance slope to increase in dense areas.
$n_{15}(x), n_{41}(x)$	Local NLoS-support fractions.	Measure how much nearby ground is geometrically shielded from the UAV.
$n_{41}(x)\log(1 + d_{2D}(x))$	NLoS-distance interaction.	Allows blocked streets to accumulate loss differently with range.
$\sigma_{\mathrm{shadow}}(x)$	Elevation-dependent shadowing proxy.	Encodes the stronger low-elevation variability reported in A2G channel models [17, 1].
$\theta_{\mathrm{norm}}(x)$	Normalized elevation angle.	Separates grazing links from near-overhead links.
$n_{41}(x)\theta_{\mathrm{norm}}(x)$	NLoS-elevation interaction.	Allows the NLoS correction to depend on whether shielding occurs at low or high elevation.
1	Bias term.	Lets each regime learn a constant offset after all physical terms are included.

Table 3.4: *Term-by-term inventory of the* `Try 78` *calibration vector.*

Each term is now described in detail.

Base masks. The ground mask is:

$$g(x) = \begin{cases} 1, & \text{if topology}(x) = 0 \\ 0, & \text{otherwise} \end{cases} \tag{3.35}$$

The building mask is $b(x) = 1 - g(x)$. The ground-NLoS support mask is:

$$n_{\text{raw}}(x) = \mathbb{1}[m_{\text{LoS}}(x) \leq 0.5] \cdot g(x) \tag{3.36}$$

This selects ground pixels that are predominantly in NLoS, i.e., pixels that are at street level and are shielded from the UAV.

Quadratic raw-prior feature. The pair $(\text{PL}^{(0)}(x)^2, \text{PL}^{(0)}(x))$ allows the linear model to fit a *quadratic* relationship in the raw prior. Including both P^2 and P as features lets the calibration curve be: $\hat{P} = \beta_1 P^2 + \beta_2 P + \cdots$, which can correct for systematic curvature in the raw prior.

Distance term. $\log(1 + d_{2D}(x))$: the logarithm of one plus the horizontal distance, providing a distance-dependent offset. The +1 prevents a singularity at $d_{2D} = 0$. The logarithm is consistent with the log-distance structure of empirical path-loss laws: a multiplicative increase in distance produces an additive increase in predicted dB loss [27, 5].

Per-pixel box-mean operator. All multiscale features use the same 2D box-mean operator BoxMean_k. For a map $f : \mathbb{Z}^2 \to \mathbb{R}$ and a window half-width $p = \lfloor k/2 \rfloor$, the value at pixel $x = (i, j)$ is:

$$\text{BoxMean}_k(f)(i, j) = \frac{1}{k^2} \sum_{i'=i-p}^{i+p} \sum_{j'=j-p}^{j+p} f_{\text{padded}}(i', j') \tag{3.37}$$

where f_{padded} is f extended outside the map boundary by *reflect* (mirror) padding: $f_{\text{padded}}(-r, j) = f_{\text{padded}}(r, j)$ and similarly at all four edges. This ensures border pixels receive a full k^2-pixel average, with the map reflected back inward rather than zero-padded.

Equation (3.37) is evaluated efficiently using a *2D summed-area table* (integral image). Let $I(i, j) = \sum_{i' \leq i, j' \leq j} f_{\text{padded}}(i', j')$ be the prefix-sum array. Then:

$$\sum_{i'=i-p}^{i+p} \sum_{j'=j-p}^{j+p} f_{\text{padded}}(i', j') = I(i+p,\ j+p) - I(i-p-1,\ j+p) - I(i+p,\ j-p-1) + I(i-p-1,\ j-p-1) \tag{3.38}$$

This reduces each per-pixel evaluation to four table lookups, making the full-map computation $O(HW)$ regardless of k. In the implementation, $k \in \{15, 41\}$ pixels, corresponding to ~15 m and ~41 m neighborhoods (1 m/pixel resolution).

Multiscale building density.

$$\rho_k(x) = \mathrm{BoxMean}_k(b(\cdot))\,(x) = \frac{1}{k^2} \sum_{x' \in W_k(x)} b(x') \tag{3.39}$$

where $W_k(x)$ is the $k \times k$ window centered at x. Since $b(x') \in \{0, 1\}$, $\rho_k(x)$ is the *fraction of pixels that are buildings* in the window. It is zero at a ground pixel surrounded only by other ground pixels, and one at a pixel deep inside a solid building block. ρ_{15} captures fine-scale building density within a ~15 m neighborhood; ρ_{41} captures a coarser ~41 m neighborhood.

Multiscale building height.

$$h_k(x) = \frac{\mathrm{BoxMean}_k(\mathrm{topology}(\cdot)\, b(\cdot))\,(x)}{H_{\mathrm{norm}}}, \quad H_{\mathrm{norm}} = 90 \text{ m} \tag{3.40}$$

This is the box mean of $\mathrm{topology}(x') \cdot b(x')$: building pixels contribute their height in metres; ground pixels contribute 0. Dividing by k^2 (inside $\mathrm{BoxMean}_k$) gives:

$$h_k(x) = \frac{1}{H_{\mathrm{norm}}\, k^2} \sum_{x' \in W_k(x)} \mathrm{topology}(x') \cdot b(x') = \frac{\rho_k(x)}{H_{\mathrm{norm}}}\, \overline{h}_{\mathrm{bldg},k}(x) \tag{3.41}$$

where $\overline{h}_{\mathrm{bldg},k}(x)$ is the *mean height of building pixels only* within the window:

$$\overline{h}_{\mathrm{bldg},k}(x) = \frac{\sum_{x' \in W_k(x)} \mathrm{topology}(x')\, b(x')}{\sum_{x' \in W_k(x)} b(x')} \quad (\text{undefined if } \rho_k(x) = 0) \tag{3.42}$$

Critically, $h_k(x) = \rho_k(x) \cdot \overline{h}_{\mathrm{bldg},k}(x)/H_{\mathrm{norm}}$, so h_k combines *both* how many buildings are present and how tall they are. A window with 50% building coverage and mean height 20 m gives the same h_k as a window with 100% coverage at mean height 10 m. The $H_{\mathrm{norm}} = 90$ m normalizer maps the joint range to approximately $[0, 1]$.

Multiscale NLoS-support.

$$n_k(x) = \mathrm{BoxMean}_k(n_{\mathrm{raw}}(\cdot))\,(x) \tag{3.43}$$

Since $n_{\mathrm{raw}}(x') \in \{0, 1\}$ (Eq. 3.36), $n_k(x)$ is the *absolute fraction of the window area* that consists of ground pixels in NLoS. Just as $h_k(x)$ conflates building coverage and building height, $n_k(x)$ conflates ground coverage and NLoS probability. We can write:

$$n_k(x) = (1 - \rho_k(x))\, \overline{n}_{\mathrm{ground},k}(x) \tag{3.44}$$

where $\overline{n}_{\text{ground},k}(x)$ is the relative fraction of *available ground pixels* that are shielded:

$$\overline{n}_{\text{ground},k}(x) = \frac{\sum_{x' \in W_k(x)} n_{\text{raw}}(x')}{\sum_{x' \in W_k(x)} g(x')} \quad \text{(undefined if } \rho_k(x) = 1) \tag{3.45}$$

Because building pixels never contribute to n_k, this feature is zero in completely open LoS areas (where $\overline{n}_{\text{ground},k} = 0$) *and* deep inside solid building blocks (where $1 - \rho_k = 0$). It reaches its maximum in narrow urban street canyons, where ground coverage is still significant but entirely shielded from the UAV.

Elevation angle.

$$\theta(x) = \arctan\left(\frac{h_{\text{tx}} - h_{\text{rx}}}{\max(d_{2D}(x), 1)}\right) \tag{3.46}$$

The $\max(\cdot, 1)$ prevents division by zero directly below the UAV. Then:

$$\theta_{\text{norm}}(x) = \text{clip}\left(\frac{\theta^\circ(x)}{90},\ 0,\ 1\right) \tag{3.47}$$

where $\theta^\circ = \theta \cdot 180/\pi$ is in degrees. $\theta_{\text{norm}} = 0$ at grazing incidence, $= 1$ directly overhead.

Shadowing proxy.

$$\sigma_{\text{LoS}}(x) = \alpha_{\text{LoS}}\,(90 - \theta^\circ(x))^{\mu_{\text{LoS}}} \tag{3.48}$$

$$\sigma_{\text{NLoS}}(x) = \alpha_{\text{NLoS}}\,(90 - \theta^\circ(x))^{\mu_{\text{NLoS}}} \tag{3.49}$$

with fitted values $\alpha_{\text{NLoS}} = 2.3197$, $\mu_{\text{NLoS}} = 0.2361$. The blended proxy is:

$$\sigma_{\text{shadow}}(x) = m_{\text{LoS}}(x)\,\sigma_{\text{LoS}}(x) + (1 - m_{\text{LoS}}(x))\,\sigma_{\text{NLoS}}(x) \tag{3.50}$$

This acts as a spatially varying uncertainty proxy: higher values at low elevation angles, where propagation conditions are more difficult.

Interaction terms. The three pairwise products $\rho_{41}(x)\log(1 + d_{2D}(x))$, $n_{41}(x)\log(1 + d_{2D}(x))$, and $n_{41}(x)\theta_{\text{norm}}(x)$ allow the linear model to capture effects where building density or NLoS support *modifies* the slope of path loss with distance or elevation angle, rather than just shifting it.

Stage 3: regime-wise linear calibration

The regime r is identified by three keys:

- *Coarse topology:* one of `open_lowrise`, `mixed_midrise`, or `dense_highrise`,
- *Propagation state:* `LoS` or `NLoS`,
- *Antenna height bin:* `low_ant`, `mid_ant`, or `high_ant`.

This gives up to $3 \times 2 \times 3 = 18$ regimes.

How topology class and antenna bin are computed. These are *per-map* scalar labels, not per-pixel quantities. They are derived once from the full topology tensor and UAV height.

The *map-level* building density and mean building height are:

$$\bar{\rho}_{\text{map}} = \frac{1}{HW} \sum_x b(x) \tag{3.51}$$

$$\bar{h}_{\text{map}} = \frac{\sum_x \text{topology}(x)\, b(x)}{\sum_x b(x)} \tag{3.52}$$

$\bar{\rho}_{\text{map}}$ is simply the fraction of map pixels that are buildings. $\bar{h}_{\text{map}}$ is the mean building height *over building pixels only* (the denominator excludes ground pixels). This is different from the per-pixel feature $h_k(x)$ in Eq. (3.41), which divides the same numerator by the full window area k^2 instead of by the number of building pixels.

The six topology classes are assigned by thresholding both quantities:

$$\text{class}_6 = \begin{cases} \texttt{open_sparse_lowrise} & \bar{\rho} \le d_1,\ \bar{h} \le h_1 \\ \texttt{open_sparse_vertical} & \bar{\rho} \le d_1,\ \bar{h} > h_1 \\ \texttt{dense_block_midrise} & \bar{\rho} \ge d_2,\ \bar{h} \le h_2 \\ \texttt{dense_block_highrise} & \bar{\rho} \ge d_2,\ \bar{h} > h_2 \\ \texttt{mixed_compact_lowrise} & d_1 < \bar{\rho} < d_2,\ \bar{h} \le h_1 \\ \texttt{mixed_compact_midrise} & d_1 < \bar{\rho} < d_2,\ \bar{h} > h_1 \end{cases} \tag{3.53}$$

with thresholds $d_1 = 0.12$, $d_2 = 0.28$ (density fractions), $h_1 = 12$ m, $h_2 = 28$ m. The three-class label (`open`/`mixed`/`dense`) is the prefix of class_6 and is used in `Try 80` reporting.

The antenna-height bin is assigned by UAV height tertiles:

$$\text{ant_bin} = \begin{cases} \texttt{low_ant} & h_{\text{tx}} \leq 58.12 \text{ m} \\ \texttt{mid_ant} & 58.12 < h_{\text{tx}} \leq 103.85 \text{ m} \\ \texttt{high_ant} & h_{\text{tx}} > 103.85 \text{ m} \end{cases} \tag{3.54}$$

The calibrated prediction for regime r is:

$$\widehat{\text{PL}}_r(x) = \mathbf{x}_{78}(x)^\top \mathbf{w}_r^{78} = \sum_{j=1}^{15} x_j(x)\, w_{r,j}^{78} \tag{3.55}$$

How $\mathbf{w}_r^{78}$ is fitted. Across all training samples whose regime label equals r, a linear regression is run:

$$\hat{\mathbf{w}}_r^{78} = \arg\min_{\mathbf{w}\in\mathbb{R}^{15}} \sum_{x\in\mathcal{S}_r} \left(y(x) - \mathbf{x}_{78}(x)^\top \mathbf{w}\right)^2 \tag{3.56}$$

where $\mathcal{S}_r$ is the set of all valid pixels in training samples belonging to regime r. This ordinary least squares problem has the closed form solution:

$$\hat{\mathbf{w}}_r^{78} = \left(\mathbf{X}_r^\top \mathbf{X}_r\right)^{-1} \mathbf{X}_r^\top \mathbf{y}_r \tag{3.57}$$

where $\mathbf{X}_r$ stacks the feature vectors and $\mathbf{y}_r$ stacks the measured path-loss values. In practice, if the matrix is near-singular (sparse regime), a fallback to a less specific regime key is used (see the fallback chain in the Try 79 section, which uses the same strategy).

The terms in Eq. (3.57) are standard linear least-squares quantities. The matrix $\mathbf{X}_r^\top \mathbf{X}_r$ is the *Gram matrix*; its diagonal entries measure the energy of each feature inside regime r, and its off-diagonal entries measure correlations between features. The vector $\mathbf{X}_r^\top \mathbf{y}_r$ measures how strongly each feature correlates with measured path loss. The inverse maps those feature-target correlations into the fitted coefficient vector $\hat{\mathbf{w}}_r^{78}$. This is the classical ordinary least-squares solution, but applied separately per propagation regime rather than globally.

An example fitted coefficient vector for the `mixed_midrise|LoS|high_ant` regime is:

$$\mathbf{w}_{\texttt{mixed|LoS|high}}^{78} = \begin{bmatrix} -0.000554,\ 0.761,\ 3.061,\ 0.127,\ 0.793,\ -2.318,\ -0.481,\ -0.157 \\ 0.155,\ -0.218,\ 0.060,\ -47.15,\ -28.08,\ -0.508,\ 48.83 \end{bmatrix}^\top$$

The large bias weight ($w_{15} = 48.83$) and the large negative shadowing-proxy weight ($w_{12} = -47.15$) partially cancel, together acting as a calibrated effective offset for this regime.

Stage 4: region replacement and clipping

$$\mathrm{PL}_{\mathrm{NLoS}}(x) = \mathrm{clip}\big(\mathbf{x}_{78}(x)^\top \mathbf{w}_r^{78}, 20, 180\big) \tag{3.58}$$

Hard clipping to $[20, 180]$ dB prevents the linear model from producing physically unreasonable values. The final hybrid path-loss prior is:

$$\mathrm{PL}_{\mathrm{prior}}(x) = \begin{cases} \mathrm{PL}_{\mathrm{LoS}}(x), & m_{\mathrm{LoS}}(x) > 0.5 \\ \mathrm{PL}_{\mathrm{NLoS}}(x), & \text{otherwise} \end{cases} \tag{3.59}$$

3.7.4 Evaluation metric: pixel-weighted RMSE

All numerical results are reported as *pixel-weighted* (pw) RMSE or MAE. Building pixels (topology$(x) > 0$) are always excluded; only valid ground pixels with a finite, physically plausible target value contribute. Let $\mathscr{V}_k$ be the set of all such valid (s, x) pairs across all evaluation samples for task k. Then:

$$\mathrm{RMSE}_k^{\mathrm{pw}} = \sqrt{\frac{\sum_{(s,x)\in\mathscr{V}_k} \left(\hat{y}_k^{(s)}(x) - y_k^{(s)}(x)\right)^2}{|\mathscr{V}_k|}} \tag{3.60}$$

This is a single square root over all pooled valid pixels, not an average of per-sample RMSEs. Units are dB (path loss), ns (delay spread), and degrees (angular spread), all in the native non-transformed domain.

3.7.5 Observed Try 78 results

Table 3.5 summarizes the path-loss RMSE values obtained from the final hybrid evaluation.

Table 3.5: *Try 78 pixel-weighted RMSE (dB) on the evaluation split. The "Role" column indicates whether the model is used as a frozen prior in* `Try 80`*, serves only as an ablation baseline, or is an intermediate construction step.*

Model	RMSE (dB)	Role in Try 80
FSPL (LoS only)	3.9540	ablation baseline (not used)
FSPL + radial residual	3.7086	ablation baseline (not used)
Coherent two-ray (LoS)	1.7516	building block → folded into Hybrid LoS
Hybrid LoS	1.7516	**frozen LoS prior** (used in Try 80)
Hybrid NLoS	3.3967	**frozen NLoS prior** (used in Try 80)
Hybrid overall	**1.9246**	combined frozen prior (Try 80 input)

Why do Coherent two-ray (LoS) and Hybrid LoS show the same RMSE? The two rows report the same value (1.7516 dB) on the *evaluation split*, but they are not identical models.

- **Coherent two-ray (LoS)** is the standalone two-ray model (Eq. 3.17) applied only to LoS pixels, with no soft clipping from an A2G envelope.

- **Hybrid LoS** applies the blend of Eq. 3.33: $0.7\,\mathrm{PL}_{\mathrm{LoS,path}} + 0.3\,\min(\mathrm{PL}_{\mathrm{LoS,path}}, \mathrm{PL}_{\mathrm{A2G,LoS}})$. For the vast majority of pixels the two-ray prediction is already below the A2G LoS envelope, so the min resolves to the two-ray value and the blend collapses to $0.7P + 0.3P = P$. The two formulas therefore coincide on almost all evaluation pixels, producing the same aggregate RMSE.

The practical difference only surfaces on pixels where the two-ray model predicts *unrealistically low* path loss (constructive-interference peaks that overshoot the measurement). On those pixels the A2G envelope acts as a physical upper-bound cap, slightly raising the prediction and preventing it from being overly optimistic. On the training set, where such edge cases appear more frequently, Hybrid LoS was marginally better than the standalone two-ray; the improvement rounds away at the reported precision for the evaluation split. The hybrid formulation was therefore kept as the frozen LoS prior in `Try 80` because it is more conservative and physically bounded, even though the difference is not visible at this precision.

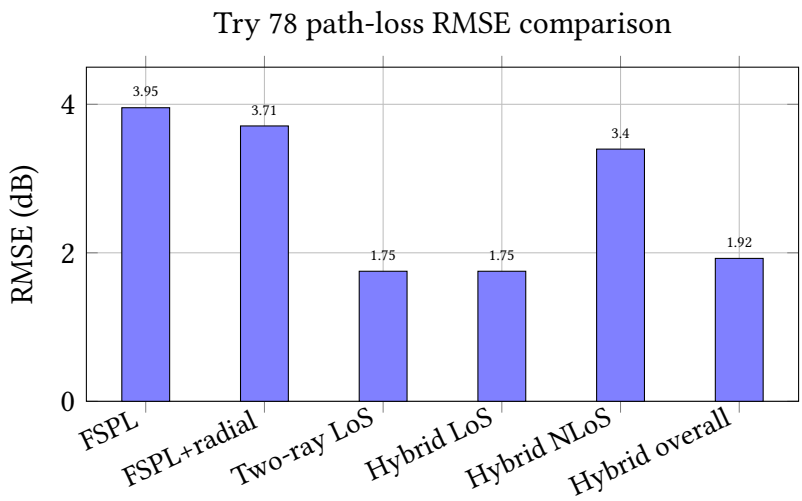

Figure 3.7: *Try 78 pixel-weighted RMSE values. The coherent two-ray model reduces LoS RMSE from 3.95 dB (FSPL) to 1.75 dB, a gain of 2.2 dB.*

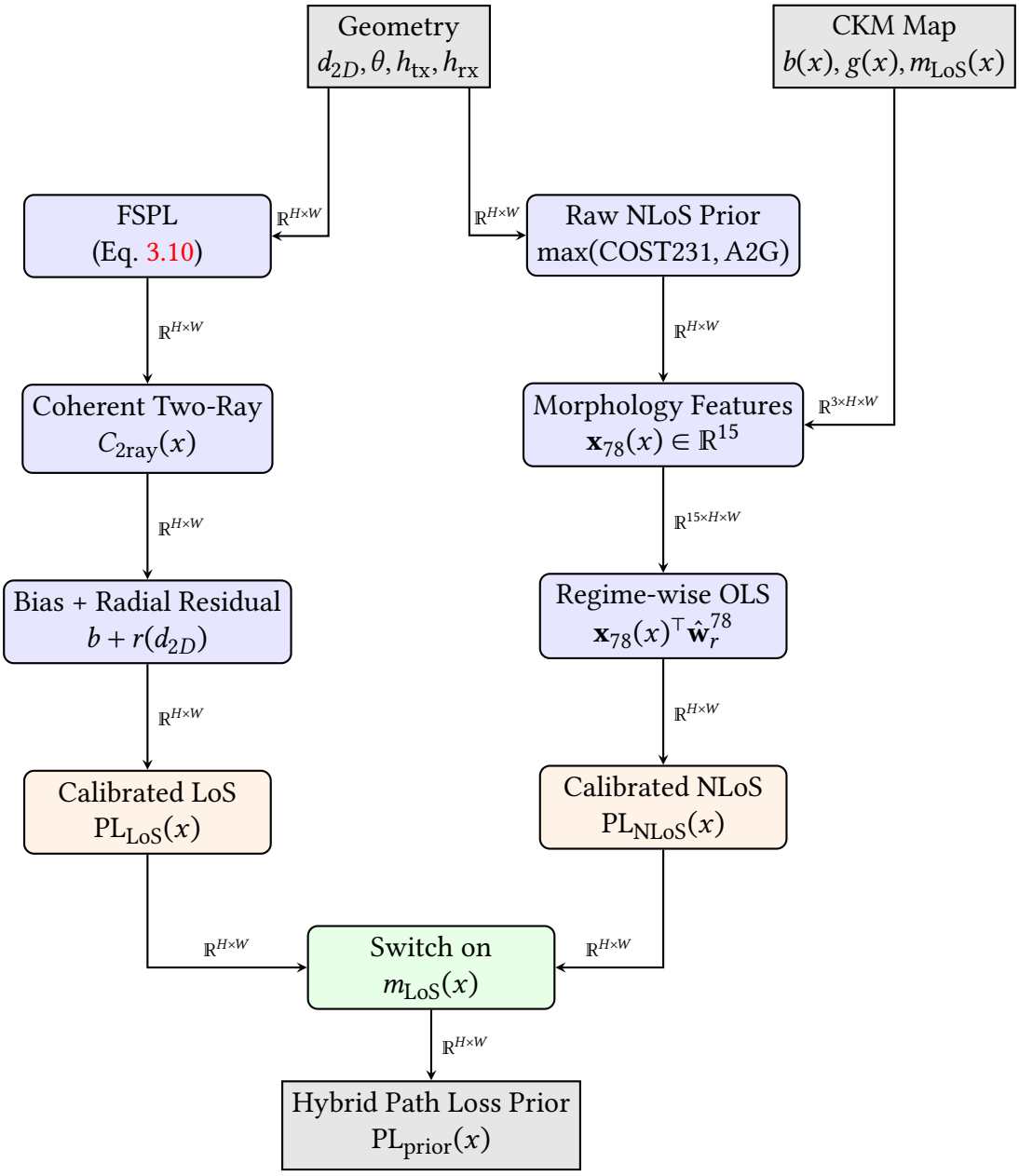

Figure 3.8: *Pipeline for the `Try 78` hybrid path loss prior. The LoS branch relies on an explicit two-ray interference model, while the NLoS branch calibrates a raw prior against map-derived morphology features using regime-wise OLS. The two branches are hard-switched based on the LoS mask to produce the final frozen prior.*

3.8 Try 79: Delay and Angular Spread Priors

3.8.1 Physical motivation for log-domain modeling

Delay spread (DS) and angular spread (AS) are strictly positive and heavy-tailed. Their empirical distributions are commonly modeled in the logarithmic domain in standard channel models [1, 18]. For a spread variable y, define:

$$z(x) = \log(1 + y(x)), \qquad \hat{y}(x) = \exp(\hat{z}(x)) - 1 \tag{3.61}$$

The addition of 1 is not cosmetic. It keeps exact zeros finite, avoids $\log(0)$, and makes the same transform usable for both delay spread in nanoseconds and angular spread in degrees. If two pixels have native spread values y_a and y_b, their difference in transformed space is

$$z_a - z_b = \log\left(\frac{1 + y_a}{1 + y_b}\right). \tag{3.62}$$

Thus, a fixed error in z corresponds approximately to a fixed multiplicative error in $1 + y$. This is more appropriate than native-space least squares for spreads because large NLoS tails would otherwise dominate the fit and the near-zero LoS mass would be ignored.

3.8.2 Pixel-wise notation for the Try 79 fit

Let n index a valid ground pixel from a training map. For a selected metric $m \in \{\mathrm{DS}, \mathrm{AS}\}$, define the native target $y_{m,n}$, the transformed target $z_{m,n} = \log(1 + y_{m,n})$, and the raw log-prior value $z^{(0)}_{m,n}$. The calibrated prior is linear in a 23-dimensional feature vector:

$$\hat{z}_{m,n} = \mathbf{x}_{m,n}^{\top}\mathbf{w}_{m,r(n)}, \qquad \mathbf{x}_{m,n} \in \mathbb{R}^{23}, \qquad \mathbf{w}_{m,r} \in \mathbb{R}^{23}. \tag{3.63}$$

The regime $r(n)$ is a discrete key containing the metric, topology class, LoS/NLoS state and UAV-height bin. In other words, Try 79 does not fit one global line through all pixels. It fits a family of small linear models, each one specialized to a propagation regime.

3.8.3 Raw spread prior in log domain

$$z^{(0)}(x) = b_{\mathrm{LoS/NLoS}}(x) + b_{\mathrm{topology}} + a_1 \log(1 + d_{2D}) + a_2\theta_{\mathrm{inv}} + a_3\rho_{41} + a_4 h_{41} + a_5 n_{41} + a_6 n_{41}\theta_{\mathrm{inv}} \tag{3.64}$$

where $\theta_{\mathrm{inv}}(x) = 1 - \theta_{\mathrm{norm}}(x)$ (high value at low elevation). Example coefficients for delay spread: $b_{\mathrm{LoS}} = 4.0$, $b_{\mathrm{NLoS}} = 11.0$, $a_1 = 0.11$, $a_2 = 0.62$, $a_3 = 0.48$, $a_4 = 0.32$, $a_5 = 0.86$, $a_6 = 0.60$.

The raw prior is not a 3GPP formula copied directly into CKM. Instead, it takes the 3GPP/WINNER lesson that spread statistics should be separated by propagation state and represented in log space [1, 18], then adds CKM-specific map morphology terms. UAV measurement studies also motivate the elevation-angle dependence: low-elevation links usually interact with more rooftops and street canyons than near-overhead links [4, 14].

Every term in Eq. (3.64) has a physical interpretation:

- $b_{\text{LoS/NLoS}}$ is the base spread level for the propagation state. NLoS starts higher because blocked paths arrive after additional reflections and diffractions.
- b_{topology} is a morphology offset. Dense and high-rise topologies receive larger offsets because they contain more reflectors.
- $\log(1+d_{2D})$ gives a smooth range dependence. Longer links have more opportunity to encounter delayed or angularly separated paths.
- θ_{inv} increases at low elevation angle, where the UAV-to-ground ray intersects more urban structure before reaching the receiver.
- ρ_{41}, h_{41}, and n_{41} describe local building density, mean building height, and NLoS support in a 41×41 window. These are direct map-derived proxies for multipath richness.
- $n_{41}\theta_{\text{inv}}$ is an interaction term. It activates when low-elevation geometry and local obstruction occur together.

3.8.4 Calibrated spread prior

Stage 2: 23-feature vector and design matrix

$$\mathbf{x}_{79}(x) = \begin{bmatrix} z^{(0)}(x)^2,\ z^{(0)}(x),\ \log(1+d_{2D}),\ \theta_{\text{norm}},\ \theta_{\text{inv}} \\ h_{\text{norm}},\ h_{\text{norm}}^2,\ \rho_{15},\ \rho_{41},\ h_{15},\ h_{41} \\ n_{15},\ n_{41},\ n_{41}\log(1+d_{2D}),\ \rho_{41}\theta_{\text{inv}},\ b_{41},\ 1 \\ c_{41},\ t_{41},\ \theta_{\text{norm}}\rho_{41},\ c_{41}\theta_{\text{inv}},\ t_{41}\rho_{41},\ t_{41}n_{41} \end{bmatrix}^\top \tag{3.65}$$

The vector in Eq. (3.65) can be grouped into five blocks:

$$\mathbf{x}_{79}(x) = \left[\underbrace{\mathbf{x}_{\text{prior}}(x)}_{\text{raw prior}},\ \underbrace{\mathbf{x}_{\text{geom}}(x)}_{\text{range/elevation/height}},\ \underbrace{\mathbf{x}_{\text{morph}}(x)}_{\text{local buildings}},\ \underbrace{1}_{\text{bias}},\ \underbrace{\mathbf{x}_{\text{int}}(x)}_{\text{interactions}} \right]^\top. \tag{3.66}$$

The first block lets the model bend the raw prior quadratically. The geometry block handles range and height. The morphology block measures whether there are nearby scatterers. The interaction block allows the model to express statements such as “density matters more at low elevation” or “being below nearby rooftops matters most when NLoS support is already high”.

Term	Mathematical role	Physical interpretation
$z^{(0)}(x)^2$	Quadratic raw-prior basis.	Allows the calibration to correct curvature in the first spread estimate.
$z^{(0)}(x)$	Linear raw-prior basis.	Keeps the calibrated model tied to the log-domain physical prior.
$\log(1 + d_{2D})$	Smooth distance basis.	Longer links can produce wider delay and angle supports because more scatterers can contribute.
θ_{norm}	Normalized elevation angle.	Separates overhead links from grazing links.
θ_{inv}	Inverse elevation angle.	Gives a large value to low-elevation links, where A2G models report stronger obstruction sensitivity [17, 1].
$h_{\text{norm}}, h_{\text{norm}}^2$	UAV-height basis and quadratic height basis.	Allows the model to learn a non-linear effect of transmitter height on spread.
ρ_{15}, ρ_{41}	Fine/coarse local building density.	Measures how many nearby reflectors and blockers exist at two scales.
h_{15}, h_{41}	Fine/coarse density-weighted building height.	Distinguishes low-rise clutter from high-rise clutter.
n_{15}, n_{41}	Fine/coarse local NLoS support.	Measures how much nearby ground is geometrically shielded.
b_{41}	Blocker-depth term.	Measures how far nearby buildings protrude above receiver height.
1	Bias term.	Gives each regime its own additive offset.
c_{41}	Transmitter-clearance term.	Measures how far the UAV is above nearby rooftops.
t_{41}	Taller-building fraction.	Counts nearby buildings whose height exceeds the UAV height.

Table 3.6: *Term-by-term inventory of the direct terms in the* `Try 79` *spread calibration vector.*

Term	Mathematical role	Physical interpretation
$n_{41}\log(1+d_{2D})$	NLoS-distance interaction.	Lets range matter differently in blocked streets than in open ground.
$\rho_{41}\theta_{\text{inv}}$	Density-elevation interaction.	Activates when dense morphology and low elevation occur together.
$\theta_{\text{norm}}\rho_{41}$	Overhead-density interaction.	Allows dense areas to behave differently when viewed from high elevation.
$c_{41}\theta_{\text{inv}}$	Clearance-elevation interaction.	Makes rooftop clearance more important at grazing elevation.
$t_{41}\rho_{41}$	Tall-building-density interaction.	Highlights dense high-rise blocks where many buildings can shadow the UAV.
$t_{41}n_{41}$	Tall-building-NLoS interaction.	Activates in blocked regions where buildings are also taller than the transmitter.

Table 3.7: *Interaction terms in the `Try 79` spread calibration vector.*

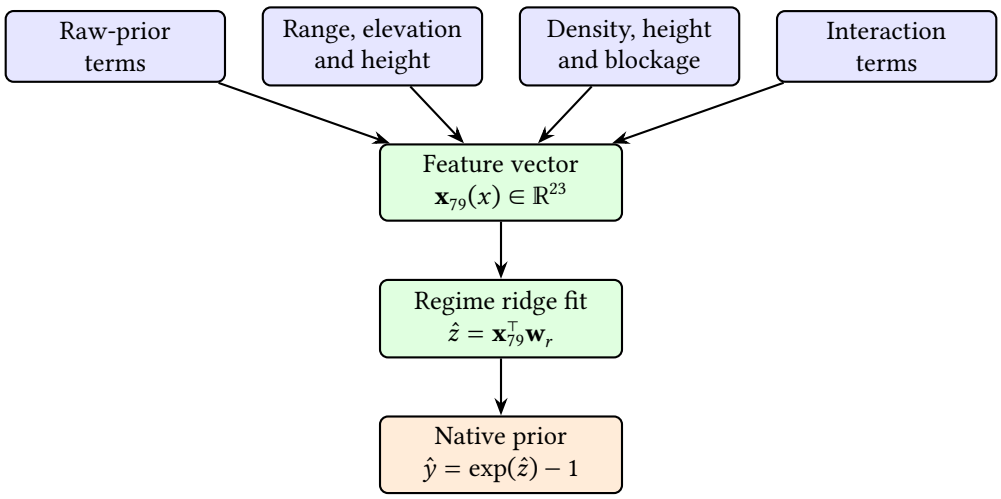

Figure 3.9: *Feature grouping used by the `Try 79` calibrated spread prior. The calibrated model is linear in $\mathbf{x}_{79}$, but the features include nonlinear transforms and interactions selected from propagation intuition.*

The new features beyond those in $\mathbf{x}_{78}$ are:

Normalised transmitter height.

$$h_{\text{norm}} = \frac{\log(1 + \max(h_{\text{tx}} - h_{\text{rx}}, 0))}{\log(401)} \tag{3.67}$$

Blocker depth.

$$b_{41}(x) = \frac{\text{BoxMean}_{41}(\max(\text{topology}(\cdot) - h_{\text{rx}}, 0)\, b(\cdot))\,(x)}{H_{\text{norm}}} \tag{3.68}$$

Measures how much buildings in the 41×41 window protrude above the receiver height. Only building pixels contribute because of the multiplier $b(\cdot)$. A tall building directly next to the receiver therefore increases b_{41}, while flat open ground leaves it at zero.

Transmitter clearance.

$$c_{41}(x) = \frac{\text{BoxMean}_{41}(\max(h_{\text{tx}} - \text{topology}(\cdot), 0)\, b(\cdot))\,(x)}{H_{\text{norm}}} \tag{3.69}$$

Measures how far the UAV is above surrounding buildings (mean altitude clearance over nearby building tops). Large c_{41} means the transmitter is well above the local skyline, which should reduce obstruction-driven delay and angular spread. Small c_{41} means the UAV is close to, or below, nearby rooftops.

Taller-building fraction.

$$t_{41}(x) = \text{BoxMean}_{41}(\mathbb{1}[\text{topology}(\cdot) > h_{\text{tx}}] \cdot b(\cdot))\,(x) \tag{3.70}$$

Fraction of buildings in the 41×41 window that are taller than the UAV. This term is zero if the UAV is above all local buildings and grows when the link is likely to be embedded in a high-rise canyon. It is especially useful for distinguishing dense but low-rise zones from dense high-rise zones.

Stage 3: regime-wise ridge regression

The regime r for Try 79 is keyed by: metric (DS or AS), topology class (6 classes), propagation state (LoS or NLoS), antenna-height bin, giving up to $2 \times 6 \times 2 \times 3 = 72$ regimes.

Valid pixels are subsampled at rate $p = 0.02$ (2%). The ridge regression:

$$\hat{\mathbf{w}}_r^{79} = \left(\mathbf{X}_r^\top \mathbf{X}_r + \lambda \mathbf{I}_{23}\right)^{-1} \mathbf{X}_r^\top \mathbf{z}_r, \quad \lambda = 10^{-3} \tag{3.71}$$

Here $\mathbf{X}_r \in \mathbb{R}^{N_r \times 23}$ stacks one row $\mathbf{x}_{79,n}^\top$ per sampled pixel in regime r, and $\mathbf{z}_r \in \mathbb{R}^{N_r}$ stacks the corresponding transformed targets. The coefficient vector $\hat{\mathbf{w}}_r^{79}$ minimizes the regularized least-squares problem

$$\hat{\mathbf{w}}_r^{79} = \arg\min_{\mathbf{w} \in \mathbb{R}^{23}} \|\mathbf{X}_r \mathbf{w} - \mathbf{z}_r\|_2^2 + \lambda \|\mathbf{D}\mathbf{w}\|_2^2, \tag{3.72}$$

where $\mathbf{D} = \mathrm{diag}(1, \dots, 1, 0)$ leaves the bias term unpenalized. The regularizer is small, but it prevents unstable coefficients when a regime contains few valid pixels or when two morphology features are strongly correlated.

Term by term, $\mathbf{X}_r^\top \mathbf{X}_r$ is the feature correlation matrix for regime r, $\lambda \mathbf{I}_{23}$ adds a small diagonal stabilizer, and $\mathbf{X}_r^\top \mathbf{z}_r$ is the feature-target correlation vector in log-spread space. This is ridge regression: it is the ordinary least-squares fit plus a penalty that discourages very large coefficients. The penalty is important here because many morphology features are related; for example, ρ_{41}, h_{41}, b_{41} and t_{41} all increase in dense high-rise areas.

Sufficient statistics are accumulated incrementally:

$$\mathbf{A}_r \mathrel{+}= \mathbf{X}_{r,s}^\top \mathbf{X}_{r,s} \quad \forall s \tag{3.73}$$

$$\mathbf{v}_r \mathrel{+}= \mathbf{X}_{r,s}^\top \mathbf{z}_{r,s} \quad \forall s \tag{3.74}$$

so the full design matrix is never materialised in memory.

Fallback hierarchy. If a regime has fewer than $4 \times 23 = 92$ pixels:

1. exact regime: (metric, topology class, LoS/NLoS, antenna bin),
2. same topology class + LoS/NLoS + *all* antenna heights,
3. same topology class + *all* LoS states + all antenna heights,
4. global topology + LoS/NLoS + all antenna heights,
5. fully global fallback: (metric, global, all-LoS, all-ant).

Stage 4: inverse transform and clipping

$$\hat{y}(x) = \exp(\hat{z}(x)) - 1, \qquad \hat{z}(x) = \mathbf{x}_{79}(x)^\top \hat{\mathbf{w}}_r^{79} \tag{3.75}$$

Regime-dependent clipping is applied after the inverse transform:

$$\hat{y}_{\mathrm{clip}}(x) = \min\big(y_{\max,r}, \max(0, \exp(\hat{z}(x)) - 1)\big), \tag{3.76}$$

with delay spread clipped to $[0, 400]$ ns and angular spread clipped to $[0, 15^\circ]$ in LoS or $[0, 90^\circ]$ in NLoS. The asymmetric angular limit is deliberate: LoS angular spread is dominated by the direct path and is physically closer to a near-zero spike plus small reflections, whereas NLoS angular spread can occupy a much wider support.

3.8.5 Try 79 results

Table 3.8: *Delay spread: calibrated test RMSE (ns) by topology class.*

Topology class	Overall	LoS	NLoS
open_sparse_lowrise	16.88	17.01	15.43
open_sparse_vertical	40.84	40.44	44.08
mixed_compact_lowrise	17.27	16.65	18.91
mixed_compact_midrise	37.88	38.26	37.25
dense_block_midrise	26.90	29.00	25.02
dense_block_highrise	43.52	40.16	45.08
Overall	**28.10**	**27.49**	**29.62**

Table 3.9: *Angular spread: calibrated test RMSE (°) by topology class.*

Topology class	Overall	LoS	NLoS
open_sparse_lowrise	10.01	10.30	6.33
open_sparse_vertical	12.70	12.98	9.97
mixed_compact_lowrise	14.37	15.99	8.17
mixed_compact_midrise	15.26	17.85	9.57
dense_block_midrise	15.53	21.20	8.17
dense_block_highrise	14.64	21.33	9.78
Overall	**13.74**	**15.28**	**8.66**

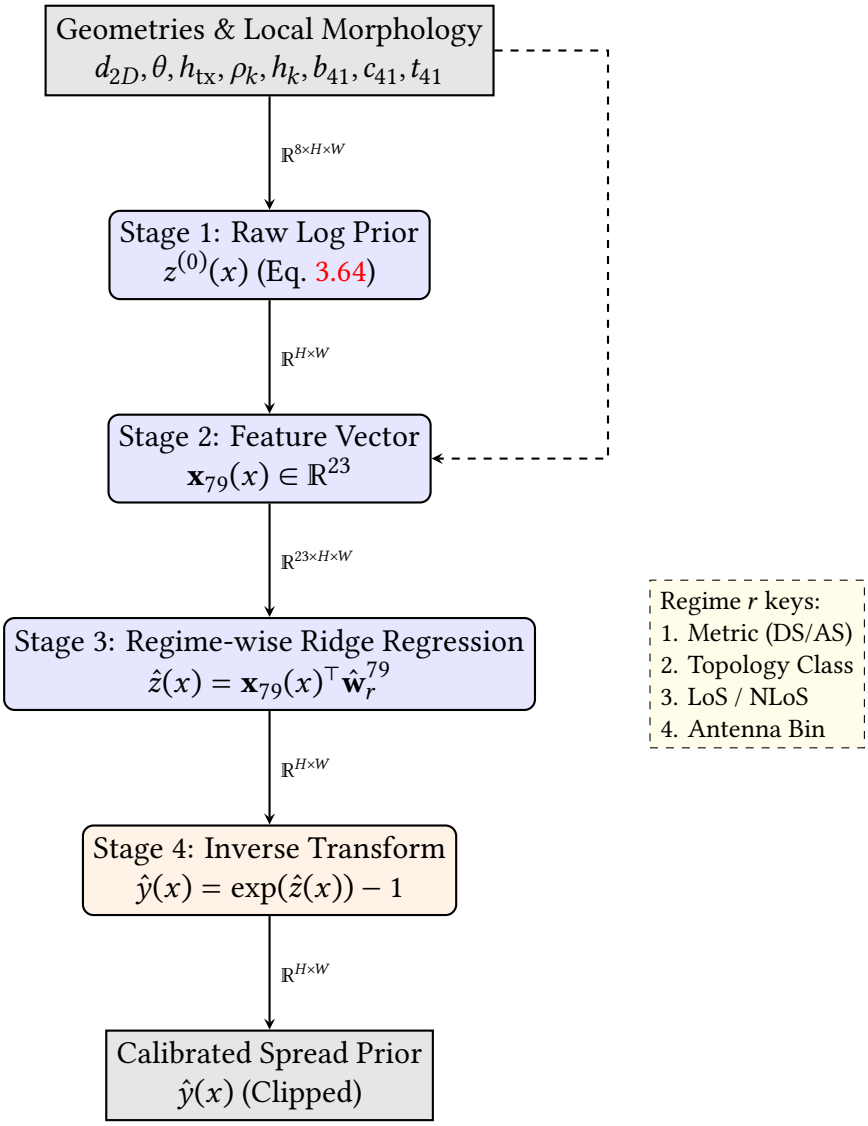

Figure 3.10: *Pipeline for the* `Try 79` *delay and angular spread priors. Both spread modalities share this identical linear architecture, predicting expected spread in the log domain via regime-wise ridge regression over 23 spatial and geometric features.*

3.9 Try 80: Joint Prior-Anchored Residual Model

3.9.1 Experimental protocol and reporting status

`Try 80` follows the same strict split philosophy as the recent distribution-first tries. The split is a city-holdout split with `split_seed = 42`, `val_ratio = 0.15`, and `test_ratio = 0.15`. Metrics and losses are computed only on valid ground pixels, using the target-specific no-data masks and the explicit LoS/NLoS masks. The model input contains nine spatial channels:

1. normalized topology,
2. LoS mask,
3. NLoS mask,
4. ground mask,
5. combined path-loss prior,
6. LoS path-loss prior,
7. NLoS path-loss prior,
8. delay-spread prior,
9. angular-spread prior.

The scalar UAV height is not appended as a constant image channel. It is encoded with a sinusoidal embedding and injected into the network through FiLM conditioning. The default `try80_joint_big` configuration uses a shared encoder-decoder with `base_width = 96`, three GMM components per task and region, α-gate bias -2, and fixed residual caps specified separately for path loss, delay spread and angular spread.

Because the full `Try 80` run is still in training, this section documents the frozen `Try 78`/`Try 79` priors and the calibration assets consumed by `Try 80`, not as final `Try 80` performance claims.

3.9.2 Prior-Anchored GMM Residual Model

`Try 80` introduces a joint multi-task architecture that predicts a *Gaussian Mixture Model (GMM)* at each pixel, rather than a simple point estimate. For each task $k \in \{\text{PL, DS, AS}\}$, region $r \in \{\text{LoS}, \text{NLoS}\}$, component $j \in \{1, 2, 3\}$, and valid pixel x, the model predicts a mixture weight, a mean and a standard deviation. Path loss is modeled in native dB space, while delay spread and angular spread are modeled in transformed $\log(1 + y)$ space for the likelihood and converted back to native units for metrics.

The native residual caps are task- and region-dependent:

$$\delta_{\max,k,r} = \begin{cases} 2\text{ dB}, & k = \text{PL},\ r = \text{LoS}, \\ 4\text{ dB}, & k = \text{PL},\ r = \text{NLoS}, \\ 30\text{ ns}, & k = \text{DS},\ r = \text{LoS}, \\ 40\text{ ns}, & k = \text{DS},\ r = \text{NLoS}, \\ 9^\circ, & k = \text{AS},\ r = \text{LoS}, \\ 13^\circ, & k = \text{AS},\ r = \text{NLoS}. \end{cases} \tag{3.77}$$

The bounded native residual is assembled from a global map-level offset and a local

pixel-level correction:

$$\Delta_{k,r,j}(x) = \mathrm{clip}\Big(\delta^{(g)}_{k,r,j} + \delta^{(\ell)}_{k,r}(x),\ -\delta_{\max,k,r},\ \delta_{\max,k,r}\Big). \tag{3.78}$$

The mean of each mixture component is then anchored to the frozen prior:

$$\mu^{\text{native}}_{k,r,j}(x) = y^{\text{prior}}_{k,r}(x) + \alpha_{k,r}(x)\Delta_{k,r,j}(x), \tag{3.79}$$

where:

- $y^{\text{prior}}_{k,r}(x)$ is the frozen regional prior from `Try 78` or `Try 79`,
- $\delta^{(g)}_{k,r,j}$ is a map-level residual offset for component j,
- $\delta^{(\ell)}_{k,r}(x)$ is a per-pixel residual refinement shared across components,
- $\alpha_{k,r}(x) \in (0,1)$ is a learned spatial gate, initialized with bias -2, so the model starts close to the prior,
- the residual is bounded to $\pm\delta_{\max,k,r}$ to prevent the DNN from deviating catastrophically from the prior.

For the spread tasks, the native anchored mean is mapped into transformed space before the Gaussian likelihood:

$$\mu_{k,r,j}(x) = \begin{cases} \mu^{\text{native}}_{k,r,j}(x), & k = \text{PL}, \\ \log\Big(1 + \max\{\mu^{\text{native}}_{k,r,j}(x), 0\}\Big), & k \in \{\text{DS}, \text{AS}\}. \end{cases} \tag{3.80}$$

The DNN also predicts per-pixel mixture probabilities $p_{k,r,j}(x)$ and component variances:

$$\sigma_{k,r,j}(x) = \sqrt{\Big(\sigma^{(g)}_{k,r,j}\Big)^2 + \Big(\tilde{\sigma}^{(\ell)}_{k,r}(x)\Big)^2}. \tag{3.81}$$

Both variance terms are constrained by sigmoid transforms to lie between $\sigma_{\min} = 0.05$ and $\sigma_{\max} = 3.0$, preventing the NLL from collapsing to unrealistically tiny uncertainty or exploding to a uselessly flat density.

The model is trained end-to-end to maximize the Negative Log Likelihood (NLL) of the measurements under this GMM. The deterministic point prediction used for RMSE evaluation is the expected value of the mixture:

$$\hat{z}_{k,r}(x) = \sum_{j=1}^{3} p_{k,r,j}(x)\mu_{k,r,j}(x), \qquad \hat{y}_{k,r}(x) = \begin{cases} \hat{z}_{k,r}(x), & k = \text{PL}, \\ \exp(\hat{z}_{k,r}(x)) - 1, & k \in \{\text{DS}, \text{AS}\}. \end{cases} \tag{3.82}$$

The final task map selects $\hat{y}_{k,\text{LoS}}$ or $\hat{y}_{k,\text{NLoS}}$ using the explicit CKM LoS/NLoS masks. This prevents a visually plausible NLoS correction from leaking into pixels whose geometric

path is LoS, and conversely prevents LoS ring structure from being used as an explanation for blocked pixels.

The training loss also includes a prior-preservation term:

$$\mathscr{L}_{\text{guard},k} = \mathbb{E}_{x\in\Omega_k}\left[\max\left(0,\ (\hat{y}_k(x) - y_k(x))^2 - \left(y_k^{\text{prior}}(x) - y_k(x)\right)^2\right)\right] \tag{3.83}$$

This *pixel-wise* prior guard penalizes the model explicitly on any pixel where its squared error exceeds the prior's squared error. The guard is not used alone; it is part of the composite optimization objective in Sec. 3.9.4.

3.9.3 Architecture: Shared UNet with FiLM

`Try 80` processes the map inputs through a shared multi-task convolutional encoder-decoder. Scalar UAV height is encoded into a condition vector $\mathbf{c} \in \mathbb{R}^{128}$ and injected dynamically into the convolutional feature maps at multiple scales via *Feature-wise Linear Modulation (FiLM)* layers.

For a given convolutional feature map $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, a linear projection of the height condition $\mathbf{c}$ outputs a scale vector $\mathbf{g} \in \mathbb{R}^C$ and a shift vector $\mathbf{b} \in \mathbb{R}^C$. The feature map is then modulated via an affine transformation:

$$\mathbf{X}' = \mathbf{X} \odot (1 + \mathbf{g}) + \mathbf{b} \tag{3.84}$$

where $\mathbf{g}$ and $\mathbf{b}$ are broadcasted spatially across the height and width.

The architecture explicitly enforces the separation of global and local effects via a dual-head design for each task:

- **Global Head (Map-Level Context)**: The deepest bottleneck feature map is globally average-pooled to produce a single vector representing the macro-context. This pooled vector, fused with the height condition, is passed through an MLP to predict the non-spatial quantities: mixture probabilities $p_{k,j}$, global offsets $\delta_{\text{global},k,j}$, and global variances $\sigma^2_{\text{global},k,j}$.

- **Local Head (Pixel-Level Refinement)**: The high-resolution output of the decoder is passed through 1×1 convolutions to predict the spatially varying fields: $\delta_{\text{local},k}(x)$, $\alpha_k(x)$, and $\tilde{\sigma}^2_{\text{local},k}(x)$.

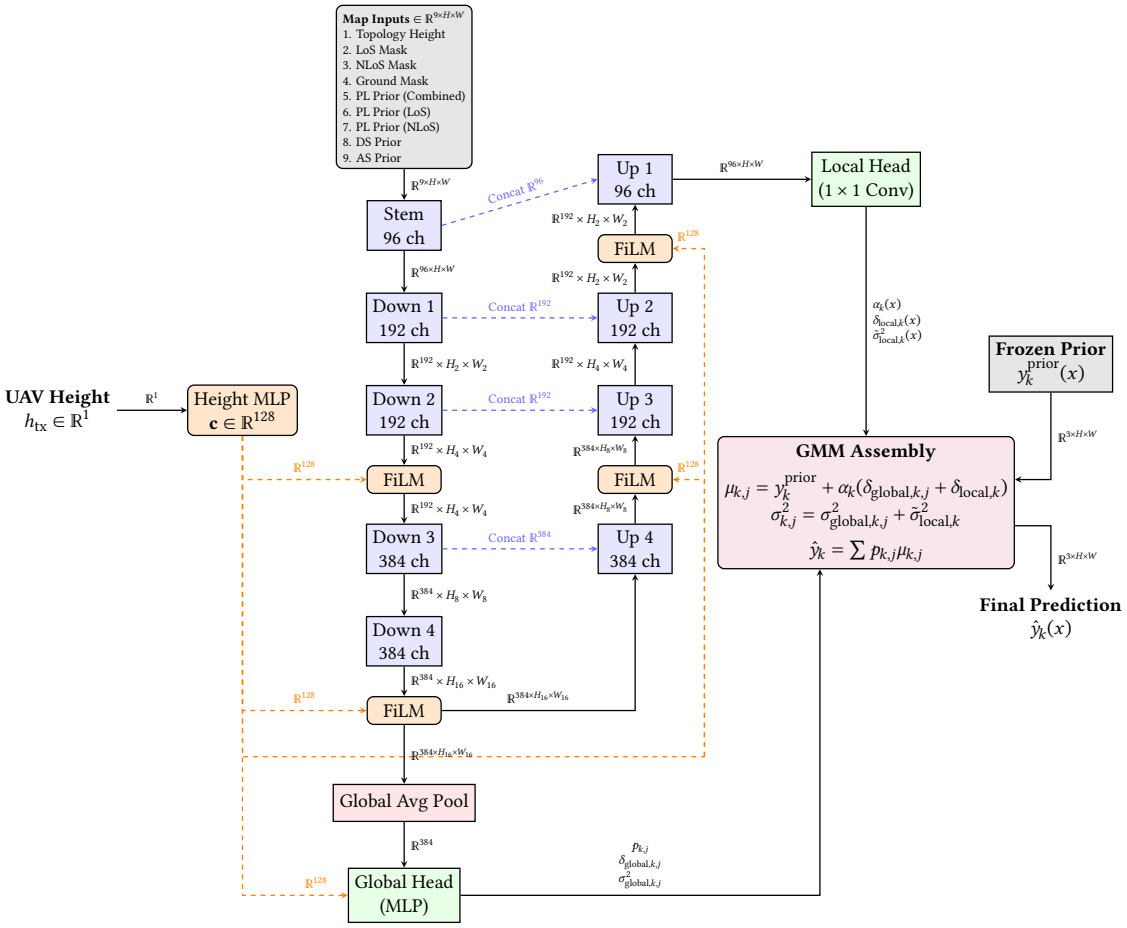

Figure 3.11: *Schematic of the* `Try 80` *Shared UNet architecture with FiLM conditioning.*

Detailed sinusoidal height embedding and FiLM path. The scalar UAV height h_{tx} is first converted into a 32-dimensional sinusoidal embedding:

$$f_i = \exp\left(\log 12 + \frac{i-1}{15}(\log 478 - \log 12)\right), \qquad i = 1, \dots, 16 \tag{3.85}$$

$$\mathbf{e}(h_{\text{tx}}) = \left[\sin\left(\frac{h_{\text{tx}}}{f_1}\right), \dots, \sin\left(\frac{h_{\text{tx}}}{f_{16}}\right), \cos\left(\frac{h_{\text{tx}}}{f_1}\right), \dots, \cos\left(\frac{h_{\text{tx}}}{f_{16}}\right)\right] \in \mathbb{R}^{32} \tag{3.86}$$

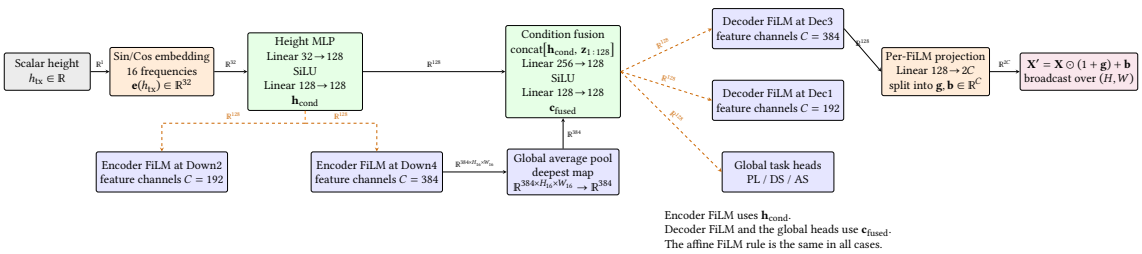

Figure 3.12: *Detailed* `Try 80` *sinusoidal height-conditioning path.*

Detailed prior-anchored GMM head. For each task k and for each region $r \in \{\text{LoS}, \text{NLoS}\}$, the model predicts a 3-component mixture. The global head outputs $\pi^{(g)}_{k,r,j}$, $\delta^{(g)}_{k,r,j}$ and $\sigma^{(g)}_{k,r,j}$, while the local head outputs per-pixel maps $p^{(\ell)}_{k,r,j}(x)$, $\delta^{(\ell)}_{k,r}(x)$, $\alpha_{k,r}(x)$, and $\tilde{\sigma}_{k,r}(x)$.

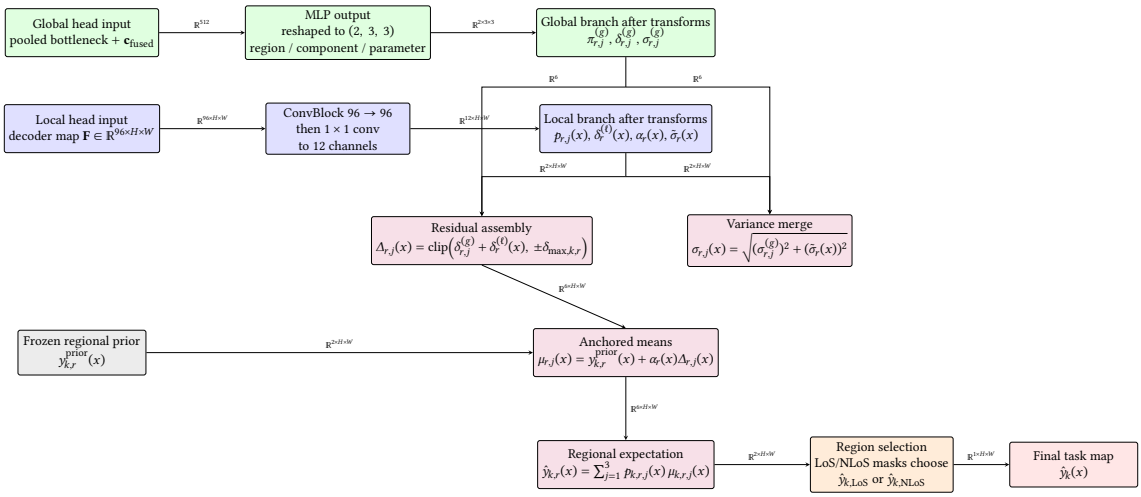

Figure 3.13: *Detailed prior-anchored GMM head used by* `Try 80`.

3.9.4 Composite Optimization Objective

The implementation utilizes a robust composite loss to stabilize the GMM training. For each task, let Ω_k be the valid ground-pixel set, let $q_k(y \mid x)$ be the predicted mixture density, and let $\hat{y}_k(x)$ be the native-space point prediction used for RMSE. The total objective is:

$$\begin{aligned}\mathscr{L} = \sum_{k\in\{\text{PL,DS,AS}\}} \Big[&\lambda_{\text{NLL}}\mathscr{L}_{\text{NLL},k} + \lambda_{\text{KL}}\mathscr{L}_{\text{KL},k} + \lambda_{\text{mom}}\mathscr{L}_{\text{mom},k} \\ &+ \lambda_{\text{anchor}}\mathscr{L}_{\text{anchor},k} + \lambda_{\text{guard}}\mathscr{L}_{\text{guard},k} + \lambda_{\text{out}}\mathscr{L}_{\text{out},k} \\ &+ \lambda_{\text{RMSE}}\mathscr{L}_{\text{RMSE},k} + \lambda_{\text{MAE}}\mathscr{L}_{\text{MAE},k} \Big].\end{aligned} \tag{3.87}$$

The numerical weights used in `try80_joint_big` are $\lambda_{\text{NLL}} = 1.0$, $\lambda_{\text{KL}} = 0.25$, $\lambda_{\text{mom}} = 0.10$, $\lambda_{\text{anchor}} = 0.05$, $\lambda_{\text{guard}} = 0.10$, $\lambda_{\text{out}} = 0.02$, $\lambda_{\text{RMSE}} = 0.40$, and $\lambda_{\text{MAE}} = 0.10$.

Map negative log-likelihood. The likelihood is computed separately in LoS and NLoS regions, then averaged over the non-empty regions. For a transformed target $z_k(x)$,

$$q_{k,r}(z_k(x) \mid x) = \sum_{j=1}^{3} p_{k,r,j}(x)\mathcal{N}\left(z_k(x); \mu_{k,r,j}(x), \sigma^2_{k,r,j}(x)\right), \tag{3.88}$$

and the regional NLL is

$$\mathscr{L}_{\mathrm{NLL},k,r} = -\frac{1}{|\Omega_{k,r}|} \sum_{x \in \Omega_{k,r}} \log q_{k,r}(z_k(x) \mid x) \,. \tag{3.89}$$

The implemented version evaluates Eq. (3.89) using $\log \sum \exp$ for numerical stability.

Distributional KL term. The global branch predicts residual components $\delta^{(g)}_{k,r,j}$. To keep those components aligned with the empirical residual distribution of the current batch, residuals $\epsilon_k(x) = y_k(x) - y_k^{\mathrm{prior}}(x)$ are softly binned into an empirical probability vector $\mathbf{u}_{k,r}$. The global mixture offsets define a predicted residual histogram $\mathbf{v}_{k,r}$. The loss is

$$\mathscr{L}_{\mathrm{KL},k} = \frac{1}{2} \sum_{r \in \{\mathrm{LoS},\mathrm{NLoS}\}} \sum_{b=1}^{B} u_{k,r,b} \log \frac{u_{k,r,b}}{v_{k,r,b}}, \qquad B = 64. \tag{3.90}$$

This term does not force a perfect pixel-wise correction. It only encourages the map-level mixture to place probability mass where the batch residuals actually live.

Moment matching. The point prediction is also constrained to have approximately the same first two moments as the target map:

$$\mathscr{L}_{\mathrm{mom},k} = \left(\bar{\hat{y}}_k - \bar{y}_k\right)^2 + \left(s_{\hat{y},k} - s_{y,k}\right)^2, \tag{3.91}$$

where bars denote masked means over Ω_k and s denotes masked standard deviation. This prevents a low NLL solution from matching local density while drifting in map-average level.

Anchor and guard terms. The anchor penalty keeps the correction small:

$$\mathscr{L}_{\mathrm{anchor},k} = \mathbb{E}\left[\alpha_k(x)^2 + 0.25\left(\delta_k^{(g)}\right)^2 + 0.50\left(\delta_k^{(\ell)}(x)\right)^2\right]. \tag{3.92}$$

The guard term is Eq. (3.83). Together, these two terms encode the design philosophy of `Try 80`: the DNN is allowed to improve the prior, but it must pay a penalty for large or harmful deviations.

Outlier budget and deterministic losses. The last mixture component is treated as the broad/outlier component. If its mean usage exceeds $\tau = 0.60$, a hinge penalty is applied:

$$\mathscr{L}_{\mathrm{out},k} = \max\left(0, \frac{1}{|\Omega_k|} \sum_{x \in \Omega_k} p_{k,\mathrm{out}}(x) - \tau\right). \tag{3.93}$$

The RMSE and MAE terms are conventional masked deterministic losses:

$$\mathscr{L}_{\mathrm{RMSE},k} = \sqrt{\frac{1}{|\Omega_k|} \sum_{x \in \Omega_k} \left(\hat{y}_k(x) - y_k(x)\right)^2}, \qquad \mathscr{L}_{\mathrm{MAE},k} = \frac{1}{|\Omega_k|} \sum_{x \in \Omega_k} \left|\hat{y}_k(x) - y_k(x)\right| . \tag{3.94}$$

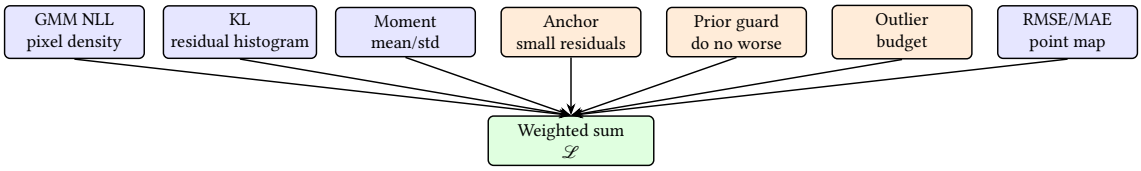

Figure 3.14: *Composite* `Try 80` *loss. The probabilistic terms train the mixture density, while the anchor and guard terms enforce the prior-residual interpretation.*

Table 3.1: *Evolution of modelling techniques and lessons across the attempt history.*

Attempts	Main technique tested	What worked	What did not work and why
1–8	U-Net and pix2pix-style cGAN for aligned image-to-image regression.	Established the full CKM pipeline and showed that dense convolutional prediction was feasible at map level.	Adversarial realism was not aligned with RMSE. It could create plausible local texture while leaving the physical propagation error unresolved.
9–14	Explicit UAV height, LoS/NLoS splits, non-finite masking, normalized height conditioning.	Made height and propagation regime visible to the model instead of hidden in the sample identity.	The network still had to rediscover the radial propagation law from pixels, which was inefficient and unstable across cities.
15–19	GroupNorm, distance-map channel, weaker or zero GAN pressure, controlled smoothing.	GroupNorm was better suited to small batches; explicit distance helped the large-scale attenuation trend.	GAN pressure mostly attacked appearance artifacts, not the hard physical errors. Too much smoothing erased real transitions.
20–22	Bilinear decoder and multiscale path-loss loss.	Try 22 became the strongest clean pre-prior path-loss baseline, with observed validation RMSE around 16.72 dB in that early contract.	The model was cleaner but still missed global radial structure and NLoS tails.
23–32	Spread-branch revisits, radial losses, support-amplitude decomposition, first prior-plus-residual ideas.	The experiments identified two separate problems: path loss needed radial physics, while spreads needed support and amplitude protection.	Attention or extra capacity alone did not solve the propagation structure. The first residual priors were conceptually promising but not yet strong.
33–40	Building-mask exclusion and restart on the newer dataset.	Established the modern evaluation rule: building pixels are invalid receiver locations and must be excluded from loss, metrics, and error maps.	Older scores became non-comparable. Some apparent progress vanished under the stricter ground-only contract.
41–50	Physical prior plus learned residual, PMNet-style backbone, prior confidence, and NLoS-tail refiner ideas.	Train-only calibration reduced a raw prior from about 67.23 dB to about 24.16 dB; Try 49 stage 1 reached about 18.87 dB against a 23.57 dB prior.	NLoS remained heavy-tailed. Several richer NLoS priors did not beat the older calibration; the best Try 50 NLoS sandbox remained around 41 dB.
51–66	PMHHNet consolidation with experts, high-frequency path, physical channels, and strict city-holdout reporting.	Try 66 reached 9.41 dB overall, with LoS already near 3.75 dB.	The remaining error was not uniform; it concentrated in NLoS regions (31.48 dB), especially low-altitude and obstructed cases.
67–75	Anti-overfitting, multi-expert variants, dual heads, heteroscedastic outputs, SWA/EMA-style training refinements.	These runs improved training discipline and made uncertainty and regime reporting more explicit.	Some configurations over-regularized the model or anchored predictions too close to the prior, so validation stalled despite more sophisticated losses.
76–77	Distribution-first GMM modelling for path loss and then spreads.	Try 76 showed that NLoS can be treated as a distributional problem; the DirectML NLoS summary reached about 3.34 dB pixel-weighted RMSE.	The method was powerful but task-specific; Try 77 provided a spread design, while final spread claims later came from the calibrated Try 79 prior.
78–80	Frozen non-DL priors for path loss and spreads, then a joint prior-anchored residual model.	Try 78 reached 1.9246 dB overall path-loss RMSE; Try 79 reached 28.10 ns and 13.74°; Try 80 unifies all three targets.	Try 80 still requires a frozen final checkpoint before its test result can be reported as the final thesis claim.

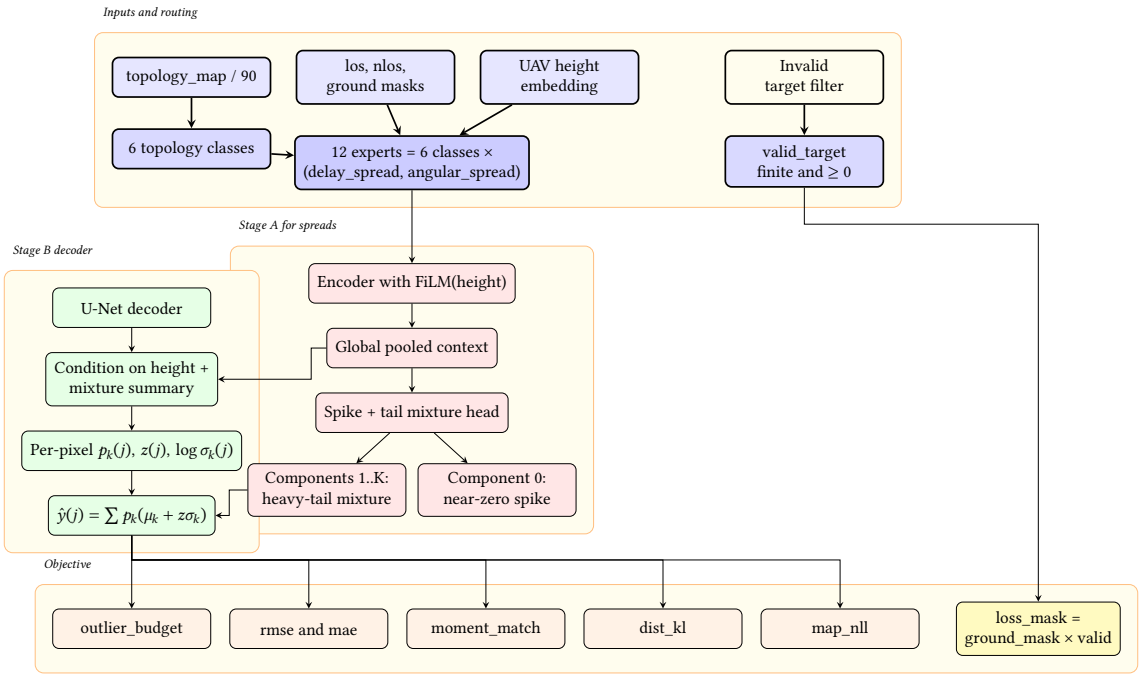

Figure 3.2: *Try 77 spread-prediction system. The Stage A mixture head separately models the near-zero spike (LoS direct path) and the heavy-tailed components (NLoS scattering). Stage B decodes these into spatially varying per-pixel predictions.*

CHAPTER 4

Results

This chapter reports the main quantitative and qualitative outcomes of the project. Final Try 80 test results are intentionally left as placeholders because the model is still being trained and the final checkpoint has not yet been frozen.

4.1 Experiments and Tests

Table 4.1 summarizes the most important milestones. The early rows are not meant as a leaderboard because masking rules and objectives changed during the project; they document why the next stage was necessary.

Try 78 is the clearest path-loss breakthrough. The coherent two-ray LoS model reduced LoS RMSE from roughly 3.95 dB with FSPL to 1.75 dB. The final hybrid prior reached 1.9246 dB overall under the pixel-weighted metric.

4.2 RMSE evolution and interpretation

The RMSE history should be read as a sequence of increasingly strict experiments, not as a perfectly homogeneous leaderboard. The dataset changed, the building-mask rule became stricter after the early experiments, and later tries used city-holdout discipline more consistently. Nevertheless, the representative values in Table 4.2 show the real direction of the project: from direct image regression, to prior-guided residual learning, to frozen calibrated priors and finally to the joint Try 80 model.

Several conclusions follow from this table. First, the project did not improve only because the neural network became larger. Some larger or more complex variants stalled when their objective was not aligned with the physical error. The biggest gains came from changing the formulation: masking invalid receiver pixels, separating regimes, injecting priors, and eventually making the prior itself strong enough to be a useful baseline.

Second, RMSE and MAE told different stories during the middle of the project. For example, Try 49 had moderate average absolute error on many samples, but RMSE

Table 4.1: *Representative experimental milestones.*

Attempt family	Main lesson	Representative result
Try 1–22	U-Net/cGAN and decoder fixes; cGAN did not solve physical error.	Try 22 became the clean pre-prior baseline.
Try 33–40	Ground-only masking became mandatory.	Building pixels excluded from loss and metrics.
Try 41–50	Prior plus residual helped LoS but NLoS remained dominant.	Try 42: 19.78 dB overall, 3.86 dB LoS, 34.47 dB NLoS.
Try 66	PMHHNet synthesis with six experts and physical channels.	9.41 dB overall, 3.75 dB LoS, 31.48 dB NLoS.
Try 76	Distribution-first path-loss modelling.	NLoS pixel-weighted RMSE ≈ 3.34 dB.
Try 78	Non-DL calibrated path-loss prior.	1.92 dB overall, 1.75 dB LoS, 3.40 dB NLoS.
Try 79	Non-DL calibrated spread priors.	28.10 ns delay; 13.74° angular.
Try 80	Joint prior-anchored model.	Validation snapshot: 1.77 dB path loss; final test pending.

remained high because a smaller number of hard NLoS cases produced very large misses. This is why the later work moved toward tail-aware and distribution-aware methods rather than only lowering the mean error.

Third, LoS and NLoS must be reported separately. Try 66 already had LoS error near 3.75 dB, while NLoS error remained above 30 dB. A single overall number would hide the fact that the LoS branch was nearly solved for that stage while the NLoS branch was still structurally wrong.

Try 79 showed that a large part of the spread targets can also be explained by geometry, morphology, and calibration without a neural network. It fitted 114 regimes in the test evaluation.

For Try 80, the current validation snapshot over 2750 samples is useful as a training diagnostic but is not the final thesis claim. It shows that the joint model can already improve some prior metrics while remaining close to the anchors. These values must be replaced by final test numbers after checkpoint selection.

4.3 Data Visualization

The most important visual diagnosis was the contrast between plausible LoS maps and failed NLoS tails. Early CNN outputs often followed the central radial trend but underestimated shadowed regions. Histogram analysis after Try 75 made this visible: the model behaved like a single-mode regressor even when the target distribution was multi-modal.

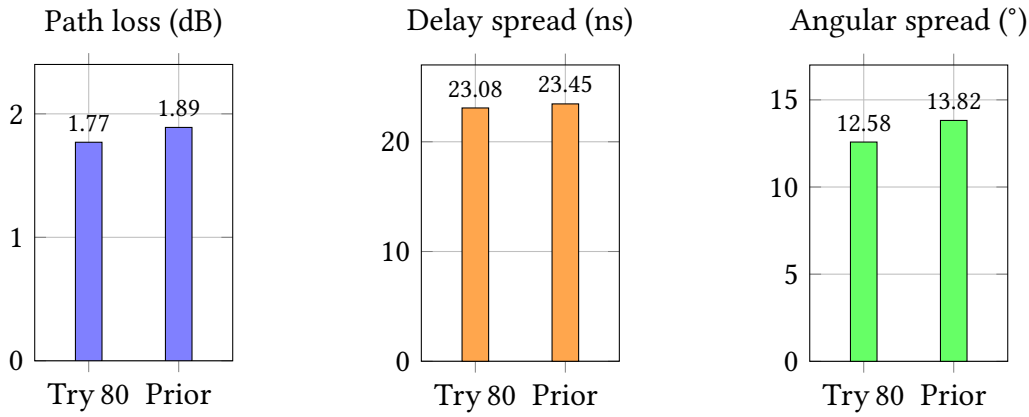

Figure 4.1: *Current Try 80 validation snapshot against its frozen priors. This figure is a placeholder until final test evaluation is complete.*

Figure 4.2 shows an example inspection panel from the Try 79 spread-prior analysis. These panels were useful for finding cases where the calibrated log-domain model worked well and cases where a topology/height regime needed a fallback.

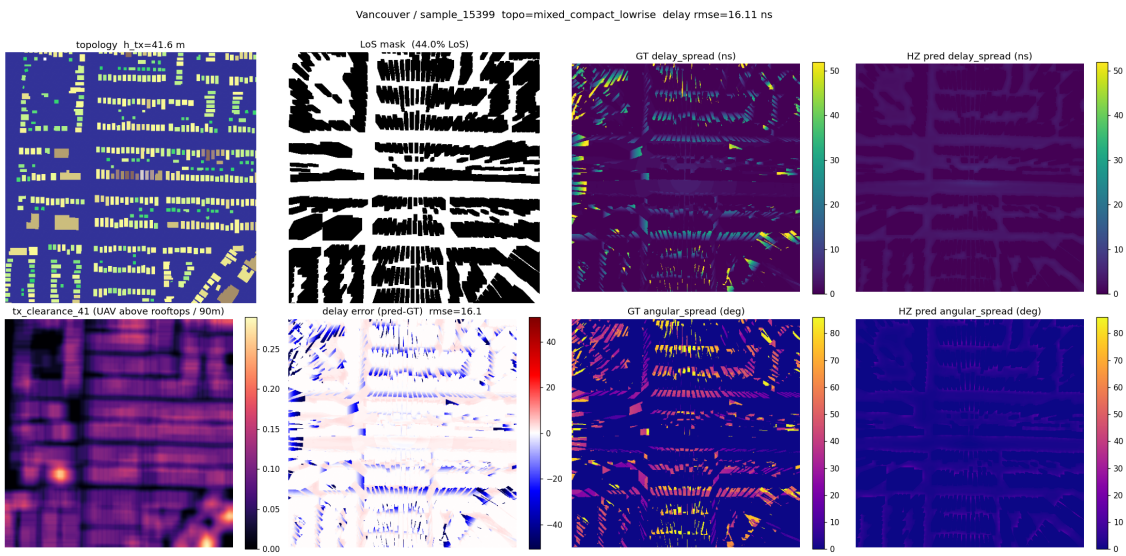

Figure 4.2: *Example Try 79 inspection panel for a Vancouver sample.*

The ongoing Try 80 run also produces a history plot comparing model metrics against

the frozen priors. It is included only as a training diagnostic until the final checkpoint and test split evaluation are frozen.

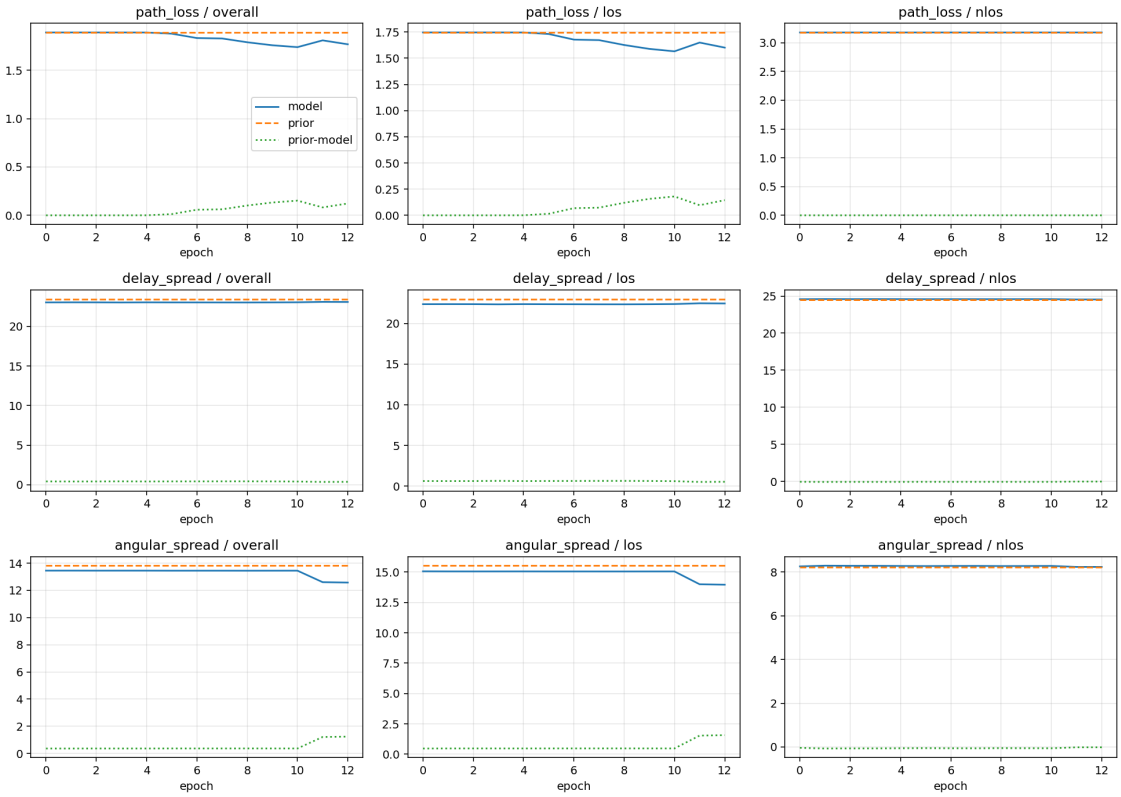

Figure 4.3: *Try 80 validation history against frozen priors. Placeholder diagnostic, not a final test result.*

The qualitative lesson from Try 78 is equally important: LoS path loss in this dataset has a strong radial ring structure. Buildings explain visibility and NLoS morphology, but in LoS they are not the main generator of the residual shape. This is why a calibrated two-ray model could outperform many neural iterations that tried to learn the same pattern from pixels alone.

4.4 Limitations

The dataset is produced by ray tracing, not by field measurements. The results therefore measure agreement with a simulator and not direct over-the-air deployment performance. The path-loss target is stored as uint8, which imposes approximately 1 dB quantization. The spread targets are also stored as integer maps: delay spread as uint16 in ns and angular spread as uint8 in degrees. A more suitable dataset for the final modelling stage would store the ray-traced targets as float16 or float32 before any display-oriented quantisation.

The effect can be written explicitly. Let $y^\star$ be the continuous ray-traced value and

$y_q = Q_\Delta(y^\star) = y^\star + \epsilon_q$ the stored value after rounding to step Δ. Under the usual high-resolution quantisation approximation, $\epsilon_q \sim \mathcal{U}(-\Delta/2, \Delta/2)$, so

$$\sigma_q = \sqrt{\mathbb{E}[\epsilon_q^2]} = \frac{\Delta}{\sqrt{12}}. \tag{4.1}$$

With $\Delta = 1$ in the physical units used here, the hidden uncertainty is 0.289 dB for path loss, 0.289 ns for delay spread, and 0.289° for angular spread. This is not the dominant error in the current Try 80 snapshot, but it is an irreducible resolution limit of the saved labels and becomes important once residual errors approach the sub-unit regime.

This also explains why getting very close to the best possible error is mathematically harder than the headline RMSE suggests. If model error and quantisation error are approximately independent, the observed squared error decomposes as

$$\mathbb{E}\big[(\hat{y} - y_q)^2\big] \approx \mathbb{E}\big[(\hat{y} - y^\star)^2\big] + \sigma_q^2.$$

Near σ_q, a real reduction in model error produces only a small change in total RMSE because of the square-root compression. In addition, many different continuous ray-tracing values collapse to the same stored integer, so the model cannot learn sub-bin structure from the labels even if the input geometry contains enough information. The heavy-tailed NLoS distributions add a second difficulty: MSE is dominated by rare large misses, so once the common LoS pixels are near the quantisation scale, further improvements depend on resolving sparse hard NLoS cases rather than on smoothing the whole map.

The final Try 80 model also still requires a frozen checkpoint and final test evaluation. The values in Table 4.5 are validation placeholders. They are useful for checking that the joint model is behaving reasonably, but the final thesis version should replace them with the completed test split report and the corresponding plots.

Finally, the current dataset fixes frequency and simulator assumptions. A general outdoor CKM model for deployment would need explicit conditioning on carrier frequency, bandwidth, antenna pattern, and possibly environment class. Though that last thing is currently covered by the city-holdout protocol, a more flexible model would need to learn to adapt to new environments from their geometry rather than just extrapolating from the training cities.

Table 4.2: *Representative RMSE evolution across the project. Values are reported under their own experiment contract and are used to show trend and diagnosis, not to claim every row is directly comparable.*

Attempt	Target / region	RMSE	Interpretation
Try 20	Path loss	≈ 17.40 dB	Bilinear decoder alone reduced artifacts but did not solve global propagation.
Try 21	Path loss	≈ 17.06 dB	Multiscale supervision helped the large-scale field.
Try 22	Path loss	≈ 16.72 dB	Best clean pre-prior baseline in the early U-Net family.
Try 41 prior	Path loss	≈ 67.23 dB	Raw physical prior was far too crude without calibration.
Try 41 calibrated prior	Path loss	≈ 24.16 dB	Regime-aware train-only calibration made the prior useful but incomplete.
Try 42	Path loss	≈ 19.78 dB	PMNet-style residual learning improved over prior-only calibration.
Try 49 stage 1	Path loss	≈ 18.87 dB	Prior-confidence and robust loss helped, but tail errors persisted.
Try 66	Path loss overall	9.41 dB	PMHHNet consolidated the physical-channel and expert approach.
Try 66	Path loss NLoS	31.48 dB	The headline bottleneck was no longer global; it was NLoS tail failure.
Try 76	Path loss NLoS	≈ 3.34 dB	Distribution-first modelling directly attacked NLoS multimodality.
Try 78	Path loss overall	1.9246 dB	Calibrated non-DL prior became the strongest path-loss baseline.
Try 79	Delay spread overall	28.10 ns	Log-domain ridge calibration met the original 50 ns project target.
Try 79	Angular spread overall	13.74°	Regime-wise calibration met the original 20 degree target.
Try 80 validation	Path loss overall	1.7692 dB	Non-final snapshot; final test result still pending.

Table 4.3: *Try 78 final hybrid path-loss prior, pixel-weighted RMSE.*

Region	RMSE	Unit
LoS	1.7516	dB
NLoS	3.3967	dB
Overall	1.9246	dB

Table 4.4: *Try 79 calibrated test results, pixel-weighted RMSE.*

Metric	Overall	LoS	NLoS
Delay spread (ns)	28.10	27.49	29.62
Angular spread (°)	13.74	15.28	8.66

Table 4.5: *Try 80 current validation snapshot. These are placeholders, not final test results.*

Target	Model overall	Frozen prior overall	Unit
Path loss	1.7692	1.8916	dB
Delay spread	23.0810	23.4494	ns
Angular spread	12.5762	13.8178	°

Table 4.6: *Approximate quantisation RMSE implied by the stored target precision. The float16 values use the unit in the upper part of the observed range (about 180 for path loss/angular spread and 678 for delay spread); float32 is effectively negligible at this scale.*

Target	Current integer step	Current floor	Float16 floor near max.
Path loss	1 dB	0.289 dB	$\approx$ 0.036 dB
Delay spread	1 ns	0.289 ns	$\approx$ 0.144 ns
Angular spread	1°	0.289°	$\approx$ 0.036°

CHAPTER 5

Sustainability Analysis and Ethical Implications

This chapter follows the ETSETB sustainability guide structure: first a sustainability matrix, then the ethical implications, and finally the relation with the Sustainable Development Goals. The analysis separates the development of the BT from the hypothetical execution of the project, because this thesis does not manufacture a physical product. Its deployable result is a software pipeline for faster CKM prediction.

5.1 Sustainability matrix

Table 5.1: *Sustainability matrix summary for the thesis.*

Perspective	Development of BT	Project execution	Risks and limitations
Environmental	GPU training, laptop use, storage, and writing.	Reduced need for repeated ray-tracing simulations.	Extra retraining, larger models, or poor reuse of cached priors could increase footprint.
Economic	Student and supervision time, plus shared HPC use.	Lower planning cost if the surrogate replaces repeated simulations.	Real deployment needs field validation and maintenance.
Social	Skills gained in wireless modelling, ML, and scientific reporting.	Faster UAV coverage planning for temporary or emergency networks.	Bias toward represented city morphologies and risk of overclaiming simulated results.

5.1.1 Environmental impact

Development of the BT. The main environmental cost of the academic work was electricity for computation. The final Try 80 line used four RTX 2080 Ti-class GPUs for approximately 12 h, consuming about 18 kWh. The final Try 76–79 stage required roughly 18 h of similar GPU time, adding about 27 kWh. The complete exploratory project exceeded 400 GPU − h; at the same approximation of 1.5 kW per four-GPU node, this corresponds to roughly 150 kWh and approximately 30 kgCO2e at an illustrative factor of 0.2 kgCO2e/kWh. This exploratory footprint is a one-time research overhead, not the marginal cost of using the final system in deployment.

Several decisions reduced avoidable impact: using city-holdout validation instead of repeated full retraining for every diagnostic, reusing calibration artifacts from Try 78 and Try 79 in Try 80, plotting from saved histories, and keeping final priors frozen rather than recomputing them blindly in every experiment.

Project execution. In operation, a trained CKM surrogate can reduce the number of ray-tracing simulations required during planning. A ray tracer is still needed to generate reliable training data and to validate new regimes, but this is a one time thing. And now this model can generalize without needing new simulations for 6G enabled 7.125GHz UAV enabled networks, which is the main future use case. The surrogate can also be used for fast what-if analyses during planning: it is not a replacement for ray tracing in every scenario, but it can evaluate many UAV heights or urban alternatives at much lower marginal energy cost.

Risks and limitations. The environmental benefit depends on reuse. If the model is retrained from scratch for every new city or frequency, the energy advantage becomes weaker, but this shouldn't be the case unless very different cities are considered. The analysis also does not include the embodied carbon of the GPUs or the data-centre infrastructure, because the project used existing shared equipment.

5.1.2 Economic impact

Development of the BT. Table 5.2 gives an indicative professional-cost view. It is not a billed project budget, but it answers the guide's request to quantify human and material resources.

Project execution. The project is economically attractive if the trained surrogate reduces repeated ray-tracing runs during network planning. The main future costs would be dataset generation for new frequencies or regions, occasional recalibration with field data, model maintenance, and quality assurance before operational use.

Risks and limitations. The largest economic risk is a simulator-to-real gap. If real measurements show that the ray-traced dataset does not transfer, additional field campaigns and recalibration would be necessary. Another risk is scope growth: making a frequency-general outdoor CKM model requires new simulations, larger storage, and more validation.

Table 5.2: *Indicative economic cost of the BT development.*

Task or resource	Quantity	Unit cost	Cost	Notes
Literature review and proposal	60 h	12 EUR/h	720 EUR	Initial scope and SOA
Dataset and baseline implementation	90 h	12 EUR/h	1 080 EUR	HDF5, U-Net/cGAN, masking
Experiment design and debugging	180 h	12 EUR/h	2 160 EUR	Tries, priors, experts, evaluation
Writing and figures	120 h	12 EUR/h	1 440 EUR	Thesis text and diagrams
Supervision meetings and his time for dataset generation from simulation	50 h	40 EUR/h	2 000 EUR	Advisor time
GPU compute, final Try 80	48 GPU-h	0.25 EUR/GPU-h	12 EUR	4× RTX 2080 Ti, 12 h
GPU compute, Tries 76–79	72 GPU-h	0.25 EUR/GPU-h	18 EUR	4× RTX 2080 Ti, 18 h
Exploratory compute	≈400 GPU-h	0.25 EUR/GPU-h	≈100 EUR	Research iterations
Electricity	—	incl. in GPU-h	0 EUR	Covered by GPU rental rate
Storage and code hosting	—	free	0 EUR	GitHub
Total indicative cost			**≈7 530 EUR**	

5.1.3 Social impact

Development of the BT. The work had a positive educational impact: it required combining propagation theory, machine learning, HPC workflows, scientific writing, and reproducible evaluation. The thesis also required continuous critical review of failed approaches, which is a useful professional practice. Inclusive language is used where applicable, and the work does not assign technical roles or expected users by gender.

Project execution. Faster CKM prediction can help plan temporary UAV coverage for public events, emergency response, rural extensions, or infrastructure restoration. The direct beneficiaries would be network planners and, indirectly, users who receive more reliable wireless coverage.

Risks and limitations. The project could negatively affect users if coverage decisions were made from overconfident simulated predictions without field validation. There is also a representativeness risk: the 66-city dataset is broad, but no dataset covers every urban morphology, construction material, or socioeconomic context. Future systems should monitor errors by city type, height bin, and LoS/NLoS regime rather than using one global RMSE as the only quality indicator.

5.2 Ethical implications

The work addresses the need for faster and more transparent radio-map prediction in UAV-enabled networks. That need was initially defined by the project proposal and

refined through experiment results: pure deep learning was not enough, so the final method keeps interpretable priors and reports regime-level errors.

The main ethical obligation is honesty about validity. The dataset is simulated through ray tracing, so the thesis must not claim real-world deployment performance without field measurements. The final Try 80 results are also left as placeholders until training finishes and a frozen checkpoint is selected.

The project aligns with professional engineering ethics by documenting assumptions, avoiding data leakage between cities, reporting limitations, and making the evaluation contract explicit. A future operational system should also include human review before coverage decisions affect public-service or emergency deployments.

5.3 Relationship with the Sustainable Development Goals

The thesis is most closely aligned with:

- **SDG 9: Industry, Innovation and Infrastructure.** The project develops a method for faster planning and evaluation of wireless infrastructure.
- **SDG 11: Sustainable Cities and Communities.** Better radio-map prediction can support temporary and resilient urban connectivity. With less needed resources to plan wireless infrastructure.
- **SDG 13: Climate Action.** The method can reduce repeated high electricity (hence carbon) costs of simulation once a validated surrogate is available, although this benefit depends on responsible reuse and avoiding unnecessary retraining.

The contribution is indirect: this thesis does not deploy infrastructure by itself, but it can reduce planning cost and help future UAV network tools make coverage decisions faster and with clearer uncertainty reporting.

CHAPTER 6

Conclusions and Future Work

6.1 Conclusions

This thesis started as a dense deep-learning surrogate for UAV radio maps and ended as a hybrid prior-anchored CKM prediction pipeline. The main lesson is that architecture alone was not enough. The decisive improvements came from understanding the target distributions, enforcing strict ground-only metrics, separating LoS and NLoS behaviour, and using physical priors where the dataset clearly followed physical structure.

The original targets were met by the finished non-DL priors for path loss, delay spread, and angular spread. Try 78 achieved 1.9246 dB overall path-loss RMSE, with 1.7516 dB LoS and 3.3967 dB NLoS. Try 79 achieved 28.10 ns delay-spread RMSE and 13.74° angular-spread RMSE on the test split. The final Try 80 neural model is still training; its final test results must be inserted once the checkpoint is selected.

The main contribution is the final synthesis: a CKM pipeline that combines calibrated propagation priors, city-holdout discipline, sinusoidal UAV-height conditioning, topology/regime diagnostics, and distribution-first residual heads. This is different from both a pure empirical propagation formula and a pure black-box CNN.

6.2 Future Directions

The most important future extension is frequency conditioning. The current dataset fixes the radio configuration, while a general outdoor CKM model should learn how path loss and spreads vary with carrier frequency, bandwidth, antenna pattern, and possibly polarization. A natural next dataset would therefore contain the same cities simulated at multiple frequencies and antenna settings.

The second extension is field calibration. The priors in Try 78 and Try 79 are already interpretable, so a small number of real measurements could be used to calibrate the simulator-to-reality gap without retraining the full model from scratch.

Finally, the Try 80 idea can be expanded with stronger global-context backbones, transmitter-oriented mixup, full corridor losses, and test-time adaptation. These should be added only after preserving the current safeguards: strict city holdout, ground-only masking, and explicit LoS/NLoS reporting.

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APPENDIX A

Reproducibility Appendix

A.1 Detailed attempt log

This appendix records the detailed experimental sequence used to build the final thesis narrative. It is intentionally more operational than the main chapters: the purpose is to show how the final design emerged from earlier failures, not to re-argue every result.

A.1.1 Early image-to-image phase

The first implementation treated CKM prediction as an aligned image-to-image translation problem. This was a reasonable starting point because the inputs and outputs share the same 513×513 grid. A U-Net generator produced dense maps and a PatchGAN discriminator judged local patches. The problem was objective mismatch. A propagation map is not a natural image: a visually plausible texture can be numerically wrong by many decibels. This is why later attempts reduced or removed adversarial pressure and moved toward explicit regression losses.

Tries 9–14 introduced height conditioning and LoS/NLoS-aware evaluation. These changes were not cosmetic. UAV height changes the path-length geometry and therefore cannot be hidden in the sample index. Likewise, LoS and NLoS pixels follow different propagation laws, so mixing them into one global score made diagnosis difficult.

A.1.2 Artifact removal and masked supervision

Tries 15–22 attacked three visible problems: checkerboard artifacts, loss of thin street and edge structure, and weak radial consistency. GroupNorm replaced BatchNorm because the full-resolution jobs used very small batches. A distance channel exposed the most important geometric variable directly. Try 20 isolated the bilinear decoder change, Try 21 isolated multiscale supervision, and Try 22 combined both. In the early contract, Try 22 was the strongest clean path-loss baseline at about 16.72 dB.

The next methodological correction was more important than another decoder change:

pixels inside buildings are not valid receiver locations. After Try 33, the project excluded $topology_map \neq 0$ pixels from the loss, from validation and test metrics, and from error-map visualization. This made the task stricter and invalidated easy comparisons to older runs that had not used the same mask.

A.1.3 Prior-guided residual phase

Try 41 made the residual formulation explicit:

$$\widehat{\mathrm{PL}} = \mathrm{PL}_{\mathrm{prior}} + r_\theta(x).$$

The initial raw formula prior was poor, with validation RMSE around 67.23 dB. Train-only regime-aware calibration improved it to about 24.16 dB. This result was simultaneously encouraging and limiting: the prior clearly contained useful information, but it was not enough to replace learning.

Try 42 moved away from the old U-Net/cGAN family toward a PMNet-style residual regressor. Try 49 added prior-confidence logic and robust loss weighting; its stage-1 branch reached about 18.87 dB, while the calibrated prior stayed near 23.57 dB. The main failure mode was a heavy tail: many pixels were moderately accurate, but some NLoS pixels produced very large errors, so RMSE stayed much larger than MAE.

Try 48 and Try 50 were useful negative evidence. Several richer NLoS priors and structural prior variants were tested, but a candidate refined prior at 24.1777 dB did not beat the older 23.5746 dB calibration, and the best Try 50 NLoS sandbox still remained near 41 dB. The conclusion was that NLoS required more than a scalar correction over a LoS-shaped prior.

A.1.4 Final distribution and prior phase

Try 66 consolidated the PMHHNet family: physical channels, GroupNorm, height FiLM, and expert routing. It reached 9.41 dB overall, but the split metric showed the deeper issue: 3.75 dB LoS and 31.48 dB NLoS. The project was no longer failing uniformly.

Try 76 reframed NLoS as a distributional problem. Instead of predicting a single value everywhere, the model predicted sample-level mixture structure and then placed components spatially. This produced a much stronger NLoS path loss result, around 3.34 dB pixel-weighted in the DirectML summary.

Try 78 then showed that LoS path loss was even more deterministic than the neural history suggested. The dominant residual over FSPL was a ring field, well explained by coherent two-ray interference with fitted effective reflection parameters. Try 79 applied a related philosophy to delay and angular spreads: use log-domain, regime-wise calibration over geometry and local morphology features. Try 80 is the synthesis: frozen Try 78/79 priors plus a neural model that predicts bounded residuals.

A.2 Try 80 configuration summary

The active Try 80 configuration uses the canonical dataset `CKM_Dataset_270326.h5`, image size 513×513, split seed 42, and validation/test ratios of 15% each after city assignment. It uses precomputed priors when available through `try80_precomputed_priors.h5`, but the priors are deterministic and can also be recomputed.

Table A.1: *Main Try 80 configuration values.*

Category	Value
Input channels	9: topology, LoS mask, NLoS mask, ground mask, combined path-loss prior, LoS path-loss prior, NLoS path-loss prior, delay-spread prior, angular-spread prior.
Height conditioning	Sinusoidal embedding with FiLM modulation.
Backbone	Four-scale GroupNorm encoder-decoder, base width 96.
Residual distribution	Three mixture components per task/region head.
Residual limits	Path loss: 2 dB LoS and 4 dB NLoS; delay spread: 30 ns LoS and 40 ns NLoS; angular spread: 9 degrees LoS and 13 degrees NLoS.
Main losses	Map NLL, distribution KL, moment matching, anchor, prior-guard, outlier budget, RMSE, and MAE.
Training	AdamW, learning rate 2×10^{-4}, weight decay 0.01, 120 epochs, batch size 1, gradient accumulation 8.

The final prediction for each task t has the same conceptual form:

$$\hat{y}_t(x) = y_{t,\text{prior}}(x) + \alpha_t(x)\Delta_t(x), \qquad 0 \le \alpha_t(x) \le 1.$$

This is the central safety mechanism. The network is allowed to improve the prior, but the residual is bounded and gated. The prior-guard term then penalizes cases where the model moves away from the prior and becomes worse.

A.3 Where the formulas are placed

The appendix is kept as an audit trail, not as a second methodology chapter. Detailed formulas are therefore placed where they are first used in the main technical narrative. This avoids having two slightly different versions of the same prior definition.

This placement makes the appendix optional for understanding the mathematics. A reader can follow the thesis from the main chapters alone; the appendix adds the chronological and reproducibility context behind those choices.

Table A.2: *Location of the main mathematical definitions.*

Topic	Location in the main text
Ground-pixel mask and RMSE contract	Chapter 3, Eqs. 3.1–3.2.
Residual-learning formulation	Chapter 3, Eq. 3.3.
Height conditioning and FiLM modulation	Chapter 3, Eqs. 3.5–3.6.
Try 78 LoS path-loss prior	Section 3.7, especially Eqs. 3.10, 3.12, 3.14, and 3.17.
Try 78 NLoS calibration	Section 3.7.3, especially Eqs. 3.27, 3.29, 3.34, and 3.58.
Try 79 spread priors	Section 3.8, especially Eqs. 3.64, 3.65, 3.71, and 3.75.
Try 80 prior-anchored GMM	Section 3.9, especially Eqs. 3.79, 3.82, and 3.83.

A.4 Final-result placeholder

The final Try 80 checkpoint is not yet frozen. Therefore, the validation values reported in the main chapters must be replaced by final test-split results before submission. The thesis should keep the current wording until that evaluation is complete: Try 80 validation results are training diagnostics, not final scientific claims.